NOFOMA 2026

15-17 June, Bergen

BI Norwegian Business School

Book of Abstracts

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Paper 2 - Work in Progress

A model of stakeholder influences on adaptation in reverse supply chains

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Keywords: Reverse supply chains; Stakeholder theory; Adaptation; Electronics; Plastics; Textiles

Purpose

This study proposes a model of stakeholder influences on reverse supply chains to explain how management plans for and responds to stakeholder constraints.

Design/methodology/approach

The Miles and Snow's adaptive cycle logic is adopted to describe the stakeholder context for electronics, plastics, and textiles in Norway. Using stakeholder theory, we examine how organizational contexts and uncertainties influence strategic decision-making and propose six conceptual propositions.

Findings

The analysis reveals that reverse supply chains adapt to contrasting environmental conditions and distinct forms of uncertainty. Textiles face an entrepreneurial problem driven by regulations, resulting in high product-market uncertainty. Plastics represent an engineering problem defined by technological uncertainty regarding sorting infrastructure and the preservation of material quality. Electronics exemplify a mature administrative problem where established extended producer responsibility (EPR) systems mitigate operational uncertainty through stabilized, business-as-usual routines.

Research implications

For managers, the model demonstrates that reverse supply chain strategies must match material maturity, shifting from innovative prospector approaches in textiles to efficiencyfocused defender structures in electronics. For policymakers, it highlights how individual rather than collective producer responsibility drives better product design and higher resource quality.

Originality/value

While research recognizes individual stakeholder constraints, scant literature compares producer responsibility organizations across distinct industrial supply chain contexts. This study bridges this gap by mapping strategic organizational typologies directly to the evolving lifecycles of reverse logistics.

Paper 3 - Full Paper

Simulating the shift: transport efficiency in electrified mass logistics

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Keywords: Mass logistics; Electrification; Rock and soil materials; Transport efficiency; Environmental impact

Purpose

While the transition to electric vehicles (EVs) is relatively straightforward in sectors with stable flows, rock and soil material transports are less predictable and often procured daily. High investment costs in vehicles and charging infrastructure make haulers hesitant without evidence of comparable or improved performance. This study investigates the effects of electrification on transport efficiency.

Research approach

The analysis employs a scenario-based simulation using operational data from diesel-driven mass transports to estimate changes in efficiency when transitioning to EVs.

Findings

Electrification significantly alters operational conditions. Approximately 60% of workdays require additional trips, typically one, to match diesel performance. Load utilization varies: 65% on days without extra trips versus 35% when additional trips occur. Energy demand frequently exceeds initial battery capacity, necessitating midday charging. However, idle periods are insufficient for required charging, highlighting infrastructure adequacy as critical. Electrification increases driving periods from two to three on most days, introducing additional mandatory breaks, thus exceeding the 9.5-hour workday limit.

Research limitations

The study is limited to Swedish conditions over 100 days. Broader geographic and temporal scopes are needed for generalization.

Practical implications

This study highlights the need for haulers to adapt operational planning to accommodate charging and additional trips.

Social implications

Electrification reduces emissions but may increase traffic volumes, posing systemic challenges.

Originality

Few studies quantify efficiency impacts and additional transport needs under electrification scenarios.

Paper 5 - Full Paper

Circular aggregate material logistics - analysis of the information constraints in the nordics

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Keywords: Circular economy; Information asymmetry; Information processing; Aggregate materials; Digitalisation; Data-value chains; Construction logistics

Purpose

This paper compares how information flows enable or hinder circularity in aggregate material logistics in Sweden and Finland. Both countries aim to reduce resource exploitation and emissions through circular economy (CE) practices yet face distinct challenges in information management. The study maps the actors in the networks, the information they exchange, and how the information flows. It also identifies barriers, enablers and governance models related to information processing.

Design/methodology/approach

The comparison draws on qualitative studies from two contexts: five Swedish research projects (2021–2025) and a Finnish project (2023–2025) both focusing on actor and resource perspectives on circular aggregate flows.

Findings

The initial findings show similarities and differences in enablers and barriers to circularity in aggregates logistics in Sweden and Finland, with timely information, traceability and collaborative planning as enablers, while clearer governance is needed. Finland emphasises data-value chains and digital platforms, whereas Sweden priorities replacing analogue routines and building trust. Research contributions The study offers insights into CE transitions in construction by showing how information and data integration shape circular material flows.

Practical implications

It provides practical suggestions for increasing digital information sharing among actors involved in circular aggregate materials.

Social implications

Improved information and data sharing reduce resource extraction, emissions, and costs, supporting EU sustainability targets.

Originality/value

This paper offers a comparative perspective on information flows in circular construction logistics, delivering actionable insights for policymakers and industry stakeholders in Nordic CE transitions.

Paper 6 - Full Paper

Coordinating reuse flows across construction projects

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Keywords: Construction supply chain; Circular economy; Cascading material flows; Supply chain planning

Purpose

Supply chain management (SCM) in construction is commonly conceptualized as a linear flow from upstream supply to on-site production. Established models describing the relationship between the construction project, construction process, and supply process have advanced understanding of coordination and planning in construction supply chains. However, these models largely assume unidirectional material flows ending at the construction site. The purpose of this paper is to extend established construction SCM models to better account for circular material flows.

Design/methodology/approach

We develop a conceptual extension of construction SCM models using cascade logic from circular economy frameworks. Cascading material flows and leakage points represent how materials exit the primary construction supply chain and enter alternative supply chains. The concept is illustrated using empirically grounded examples of surplus gravel masses and residual wood materials from construction contexts.

Findings

The analysis shows that introducing cascading flows challenges the implicit linearity of established construction SCM models. Surplus gravel masses and residual wood materials may generate parallel supply chains with distinct actors, planning horizons, and coordination requirements, complicating traditional planning assumptions.

Research limitations/implications

The paper is conceptual, and the empirical material is illustrative. Future research should conduct extended empirical studies on how cascading material flows can be systematically integrated into construction supply chain planning and performance measurement.

Originality/value

The paper contributes by questioning linear supply chain assumptions in construction SCM and proposing a cascade-based conceptual extension that supports the integration of circular material flows into construction logistics and supply chain planning.

Paper 9 - Full Paper

Business models in cycle logistics: a typology of operators

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Keywords: Cycle logistics; Business models; Urban logistics; Qualitative research; Sustainability

Purpose

Cycle logistics has emerged as a response to urban freight challenges, combining environmental performance with operational efficiency in dense city centers. While prior research has documented its environmental benefits, the diversity of business models underpinning this sector and the conditions of their economic viability remain underexplored. This paper addresses this gap by developing an empirically grounded typology of cycle logistics business models.

Design/methodology/approach

The study relies on 33 qualitative case studies of cycle logistics operators. Data were collected through semi-structured interviews with founders, managers, dispatchers, and experienced couriers, complemented by on-site observations. Interviews were transcribed and analyzed using thematic coding with NVivo. The analysis combined intra-case analysis, cross-case comparison, and pattern identification to construct a business model typology.

Findings

Six ideal-typical business models are identified: integrated logistics service providers, waste and resource recovery operators, segment-focused expert operators, massified generalist operators, local multi-activity operators, and social insertion-based enterprises. These models differ in value propositions, revenue structures, operational organization, and relationships with public authorities. The analysis also highlights path dependencies, hybridization dynamics, and shared vulnerabilities related to regulation, access to urban logistics space, and margin pressure.

Originality/value

This paper provides one of the first empirically grounded typologies of cycle logistics business models. It contributes to logistics and supply chain management literature by clarifying the organizational and economic diversity of cycle logistics and by offering a framework to support academic analysis, managerial decision-making, and public policy in the transition toward low-carbon urban logistics.

Paper 10 - Work in Progress

A multiple case study to investigate decision making in reverse logistics

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Keywords: Circular economy; Reverse logistics; Decision making; Centralization and Decentralization

Purpose

The purpose of this study is to investigate how contextual factors influence reverse logistics (RL) strategy and the centralization and decentralization of facilities.

Design/methodology/approach

An exploratory multiple-case study approach is employed, comprising three cases drawn from two Original Equipment Manufacturers (OEMs). Data were collected through 14 semi-structured interviews, complemented by secondary documents. Cross-case analysis integrates empirical findings with theory.

Findings

The results show that product characteristics, particularly complexity, customization, value, lifecycle stage, usage condition, and return volume, are central drivers shaping different RL strategies. These product characteristics translate into decision criteria related to quality control, economic sustainability, and in-house knowledge development. In addition, After-market strategy influences in-house knowledge, while sales strategy affects the geographical dispersion of sold products and collaboration among actors. These decision criteria, in turn, impact the RL strategy and decisions regarding the centralization or decentralization.

Research limitations/implications

By focusing on refurbishment, the study is bound to one R-strategy, allowing observed differences to be attributed to contextual factors rather than to variation in R-strategies.

Practical implications

The results highlight the need for OEMs to adopt a holistic view when establishing RL, offering guidance for both researchers and practitioners.

Originality/value

The study contributes to the RL and circular economy literature by providing empirically grounded insights into how contextual factors shape OEM's RL decision-making. While research has examined RL, less is understood about how contextual factors influence RL strategy and the centralization or decentralization of facilities.

Paper 11 - Work in Progress

Managing knowledge sharing for improved preparedness: developed countries' readiness

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Keywords: Knowledge management; Emergency preparedness; Civil society actors; Theoretical development

Knowledge management (KM) is recognized as a vital element in enhancing the readiness of developed countries (Kaneberg et al., 2016). Although various civil society actors (CSAs) are considered essential resources in preparedness, as noted by Kaldor (2001), the question of how KM can improve the efficiency and effectiveness of CSAs has received limited attention in current humanitarian logistics research. This paper aims to analyse the KM's role in strengthening the CSA's ability to respond to complex crises and wars

Design/methodology/approach

This is a conceptual paper that focuses on developing KM-relevant models in the field of humanitarian logistics. It places particular emphasis on understanding KM's potential in the field of humanitarian logistics and proposes a conceptual model that can be used to organize research in the more specific area of network preparedness among CSAs. We examine this from a KM perspective and explore how models from KM could be integrated into the preparedness planning field to better facilitate CSA in addressing all types of complex crises in developed countries.

Findings

We explore the latest research developments and propose future research directions on this promising topic

Research limitations/implications

The paper offers several propositions and suggestions for further research in the field, but it does not include original research.

Originality/value

The paper reveals a significant gap in the theoretical foundation of humanitarian logistics, particularly in the preparedness literature, where KM has been somewhat studied as a crucial element; however, the theoretical connections to preparedness have not been explored sufficiently

Paper 12 - Full Paper

Cybersecurity supplier development: a paradox lens

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Keywords: Information security; Capability; Strategy; Performance; Technology

Purpose

The paper sheds light on how companies that practice supplier development (SD) manage cybersecurity in development-oriented activities with suppliers.

Design/methodology/approach

The paper is based on qualitative interviews with four large Danish manufacturers that all practice SD and are sampled along two dimensions: (1) regulatory pressure to comply with NIS2 or not, and (2) whether cybersecurity is integrated into the SD program or not.

Findings

The results show that regulatory cybersecurity requirements such as NIS2 drive concrete efforts to focus on supplier cybersecurity. The study also indicates that integrating cybersecurity into supplier relationship management is a significant organizational task, and that new supplier segmentation models could be developed as a supplement to the models used to select suppliers for SD programs.

Research limitations/implications

The paper is based on the practices in four large Danish manufacturing companies; therefore, the results cannot be generalized.

Research implications

The paper challenges the traditional dogma regarding the criteria for selecting suppliers in which companies choose to invest in SD. Regardless of how suppliers are segmented, there is a risk of supplier-induced data compromise.

Practical implications

Companies should expand SD to include cybersecurity, not just performance. NIS2-driven compliance alone is insufficient; a tailored approach based on supplier risk and digital maturity is needed. Cross-functional governance and collaboration are key. While based on four cases, the study offers actionable insights for managing supplier-induced cyber risks.

Originality/value

To the authors knowledge the paper is the first to study cybersecurity in the context of SD.

Paper 13 - Full Paper

The status-quo of cross-sector partnerships in humanitarian logistics

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Keywords: Cross-sector partnerships; Governance; Social exchange theory; Humanitarian logistics

Purpose

With the growing complexity in the humanitarian sector, cross-sector partnerships (CSPs) in humanitarian logistics (HL) leverage collaborative efforts to effectively and efficiently address humanitarian needs. However, CSPs are still fragmented and inefficient that costs not only resources but also human lives. This study aims to present the building blocks of CSPs in HL. It also outlines the barriers to and opportunities for enhancing CSPs in humanitarian contexts. Theoretical framework This research incorporates the "governance" concept to explain the regulation and coordination of CSPs. In addition, social exchange theory (SET) offers an instrument to explore the behavioural dynamics in CSPs as a social construct.

Design/methodology/approach

An explorative empirical study using semi-structured interviews are conducted with practitioners in HL representing three key actors in CSPs – public, private, and non-profit sectors. Thematic coding is used for qualitative data analysis.

Research implications

Utilising the "governance" concept and SET, the study examines the role of CSPs in HL through a systemic and interdisciplinary perspective. This research brings together the best of both worlds – "governance" from public management and SET from organisations studies – to leverage the theoretical rigor of humanitarian operations research. Societal and practical implications This study emphasises the role of CSPs in strengthening humanitarian responses and inter-organisational partnerships. Identified barriers and drivers for CSPs in HL also support decision-makers to reflect on the practices of partnering and to establish more effective partnerships in the future. Thus, the research offers insights for strengthening community resilience and solidarity in crises.

Paper 14 - Work in Progress

Resilience in pharmaceutical supply chains - scenario analysis

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Keywords: Pharmaceutical supply chains; Supply chain resilience; Scenario-based approach.

Purpose

Pharmaceutical supply chains are exposed to systemic disruptions driven by geopolitical instability and environmental threats. This paper conceptualises resilience in pharmaceutical supply chains through a scenario-based approach that captures the interaction between demand and supply shocks under varying external conditions.

Design/methodology/approach

The paper develops scenarios based on prioritised driving forces and combinations of demand volatility and supply disruption, shaped by geopolitical and environmental drivers. Scenarios are based on secondary data analysis and interviews of supply chain actors. The scenarios are used as an analytical lens to explore vulnerabilities, adaptive capacities, and response strategies across pharmaceutical supply chains.

Findings

Scenario analysis provides a structured way to examine how different combinations of demand shifts and supply disruptions may unfold under varying geopolitical and environmental conditions. Geopolitical and environmental risks often interact, amplifying disruption and constraining recovery options. The four developed scenarios reveal diverse resilience challenges that require distinct resilience capabilities.

Research limitations/implications

Future research could empirically test the scenarios across different pharmaceutical products, regions, or regulatory contexts.

Practical implications

The scenarios offer decision-makers a structured way to stress-test pharmaceutical supply chains and to design targeted resilience strategies, including diversification, strategic stockpiling, and regulatory coordination.

Social implications

By addressing access to essential medicines under disruption, the paper informs debates on public health security, equity, and policy preparedness.

Originality/value

The paper contributes to supply chain resilience research by using scenarios as a forward-looking approach as opposed to retrospective case studies, making it more appropriate for a complex, fast-changing environment.

Paper 15 - Work in Progress

A conceptual decision-making framework for reverse logistics

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Keywords: Circular economy; Reverse logistics; Conceptual framework; Reverse logistics strategy

Purpose

The purpose of this study is to develop a conceptual framework to support original equipment manufacturers (OEMs) in adopting reverse logistics (RL) strategies within circular supply chains. The study aims to identify key contextual factors influencing the adoption of RL strategies and to explore the challenges OEMs face when implementing different RL strategies. Accordingly, the study addresses two research questions: (1) Which supply chain contextual factors influence OEMs' strategic choices in developing RL? and (2) What challenges arise in the adoption of RL strategies?

Design/methodology/approach

The study adopts a qualitative research approach, combining a literature review with a case study. Empirical data were collected through 14 semi-structured interviews, complemented by secondary documents provided by the case company.

Findings

The findings indicate that the RL strategy is significantly influenced by its interaction with forward supply chain strategies. Key contextual factors influencing RL strategy include logistics strategies, marketing and sales channels, after-market services, and product characteristics. The findings further reveal that inconsistencies in the product's quality, pricing, and response times are the challenges that OEMs consider in RL strategy adoption.

Research limitations/implications

The study is based on a single case study, which may limit generalizability.

Practical implications

The proposed framework supports OEMs in adopting appropriate RL strategies.

Social implications

By enabling effective RL, the framework supports waste reduction, resource efficiency, and circular economy

Purpose

Originality/value

The study contributes a novel conceptual framework linking forward supply chain strategies to RL decision-making, offering new insights into circular supply chain decision-making.

Paper 16 - Full Paper

Robustness, flexibility and preparedness in supply chains: an information processing perspective

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Keywords: Supply chain risk; Robustness; Flexibility; Preparedness; Resilience

Purpose

This paper examines how different types of supply chain risk exposure are associated with firms' preparedness through two distinct mechanisms: robustness and flexibility. While both concepts have been discussed as dimensions of supply chain resilience, empirical evidence on how specific risk characteristics affect these alternative preparedness responses is limited.

Design/methodology/approach

The study tests a model where supply chain risk exposure influences preparedness indirectly via robustness and flexibility. Risk exposure distinguishes recurrent, high-probability operational risks and low-probability, high-impact disruption risks. Robustness and flexibility are operationalized as distinct but complementary capability constructs. The empirical analysis is based on survey data collected from Finnish manufacturing and trading firms between November 2025 and January 2026.

Findings

Different risk categories are associated with different preparedness responses. High-probability operational risks are more strongly linked to robustness-oriented solutions emphasizing stability, redundancy and resistance to change. In contrast, low-probability, high-impact risks are more strongly associated with flexibility-oriented solutions that enable rapid adaptation and reconfiguration. Both mechanisms contribute positively to overall preparedness.

Research limitations/implications

Limitations of the study include the focus on Finnish firms and the reliance on survey data. Future research could use longitudinal designs and extend the model to examine performance and recovery outcomes.

Practical implications

The results suggest that firms should align preparedness investments with the nature of risks faced, rather than rely on uniform resilience strategies.

Originality/value

The paper advances supply chain risk management and resilience research by empirically demonstrating that robustness and flexibility represent distinct preparedness mechanisms associated with different risk exposure profiles.

Paper 17 - Work in Progress

Enabling supply chain resilience in manufacturing SMEs

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Keywords: Supply chain resilience; Supply chain disruptions; Small and medium enterprises (SMEs); Resilience capabilities of SMEs; Enablers and challenges to resilience

Purpose

The purpose of this paper is to identify capabilities, enablers and challenges for small and medium enterprises (SMEs) in the manufacturing sector to become resilient to supply chain disruptions.

Design/methodology/approach

This paper is based on a structured mapping review of the literature from peer reviewed journals indexed in Scopus and Web of science, focusing on supply chain resilience in the context of SMEs.

Findings

The preliminary findings show that SMEs differ from other organizations in terms of how they are affected by supply chain disruptions and how they respond to disruptions to be resilient. The paper also illustrates how SMEs leverage their existing capabilities to manage disruptions and become resilient. Research limitations/implications (if applicable) This is a conceptual paper based on the review of literature and needs empirical validation. Practical implications (if applicable) This paper contributes to supporting SMEs to become resilient to supply chain disruptions by building on their existing capabilities. The idea of the paper is to provide actionable insights to manage supply chain disruptions, based on the findings of the literature.

Originality/value

This paper focuses on supply chain resilience in the SME context by emphasizing their susceptibility to supply chain disruptions and also highlights how SMEs leverage their existing capabilities to manage disruptions and become resilient. This paper also addresses the additional support which the SMEs require to manage supply chain disruptions and become resilient.

Paper 18 - Full Paper

Building resilience in SME supply chains : empirical insights from the industrial wood house industry

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Keywords: Supply Chain Disruptions; Supply Chain Resilience; Small and Medium Enterprises (SMEs); Industrial Wood House Supply Chain; Resilience in SMEs

Purpose

The purpose of this paper is to describe the impact of supply chain disruptions on small and medium enterprises (SMEs) in the industrial wood house industry supply chain and how they respond to supply chain disruptions.

Design/methodology/approach

The paper is based on the empirical findings from industrial wood house industry supply chain. The data was collected from observations, company documents and semi structured interviews. Pattern matching will be used to identify the specific risks and resilience capabilities in SMEs from industrialised wood house industry supply chain.

Findings

The preliminary findings depict the supply, demand and process risks induced by the supply chain disruptions on SMEs within industrial wood house industry. The paper also describes how these SMEs leverage their existing capabilities and respond to supply chain disruptions and become resilient.

Research limitations/implications

This study is conducted within a specific industry and region and hence the findings may have limited generalizability. However, the findings may be applicable to manufacturing SMEs with similar challenges.

Practical implications

This paper increases risk awareness among SMEs within the industrial wood house industry supply chain and explores current strategies deployed by SMEs to become resilient. The idea is to provide guidelines to SMEs to manage disruptions and become resilient by leveraging their existing capabilities.

Originality/value

This paper contributes to identifying the supply chain risks experienced by SMEs in a new context and strategies to become resilient based on the knowledge of current strategies deployed by SMEs in this context.

Paper 19 - Work in Progress

Geopolitical disruptions and global supply chains: laying theoretical foundations and building conceptual framework.

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Keywords: Supply chain disruptions; Supply chain design; Geopolitics; Theory building; International relations

Purpose

This conceptual paper develops a robust theoretical framework for Geopolitical Supply Chain Disruptions (GSCD) by integrating and synthesizing extant Operations Management (OM) and Supply Chain Management (SCM) literature with a novel, interdisciplinary perspective from International Relations (IR). It aims to enhance explanatory power and overcome the limitations of conventional theories in understanding GSCD, providing a novel lens to examine the effects of escalating geopolitical tensions on supply chain design, performance and resilience.

Design/methodology/approach

The study identifies theories routinely applied in OSCM and their limitations in analysing the GSCD phenomenon. Building on a Practical Theorising Model, the study integrates key IR theories: Realism, Liberalism, and Constructivism to develop a conceptual framework that bridges SCM/OM and IR perspectives.

Findings

The study presents theoretical propositions addressing the causes, propagation and consequences of geopolitical disruptions in supply chains. These propositions combine theories routinely used in SCM/OM literature with IR perspectives, laying a foundation for new managerial strategy and future empirical validation. The study also offers a formal definition of GSCD, useful for future researchers.

Originality/value

This study is the first to theorize geopolitical disruptions in supply chains through an IR lens, offering an innovative interdisciplinary approach. It contributes a novel conceptual framework and research directions for advancing this still emerging but critical field.

Paper 20 - Full Paper

Auditable end-of-life textile collection for extended producer responsibility

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Keywords: Textile waste collection; MCDA; Extended producer responsibility; Digital traceability; Reverse logistics

EU-wide textile EPR is approaching, yet collection performance varies widely and evidence is fragmented. This paper asks how textile waste collection systems should be designed and selected so that they (i) deliver circularity outcomes (capture, quality, reuse/recycling), (ii) remain feasible and scalable across contexts, and (iii) support EPR target-setting, fee modulation, and emerging digital traceability requirements.

The study develops and applies a multi-criteria decision analysis (MCDA) framework combining a systematic literature review with a case-based, mixed-method empirical phase. Primary data consists of semi-structured practitioner/expert interviews supported by an optional pre-survey providing quantitative anchors (e.g., volumes, contamination, cost). Qualitative material is analysed in NVivo using hybrid coding (deductive MCDA criteria and inductive themes) and triangulated with workshops and provided additional documents. Findings are synthesized into archetype-level performance signals and contextual suitability patterns, interpreted through a socio-technical transition lens.

No single "best" model emerges. The most persistent trade-offs are capture versus quality, service level versus cost, and traceability versus proportionality. Bring-based infrastructure often provides scalable capture, while higher-service channels can protect quality but become financially binding. Data maturity is uneven, constraining enforceable targets and differentiated fees.

Small-N, context-specific cases limit statistical generalization; the contribution is a transparent, transferable evaluation logic.

The paper derives "if-then" rules for differentiated EPR obligations/fees and proposes minimum viable reporting requirements to make targets auditable.

Supports implementable textile EPR that increases reuse/recycling, reduces leakage, and improves transparency and equity.

Integrates collection-system MCDA with EPR design levers and digital innovation readiness in one operational decision framework.

Paper 21 - Full Paper

Under pressure: how institutional forces shape the transition towards green and resilient transport supply chain

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Keywords: Sustainable transport; Green logistics practices; Climate resilience; Institutional theory; Decarbonized robust freight

Purpose

Inter-organizational cooperation is an important tool for green and resilient transportation systems in response to climate change. Such measures can be facilitated or hindered by different external pressures. The study investigates how transport supply chain actors are affected by coercive, normative, and mimetic institutional pressures in their actions toward reducing climate impact from, and improving climate resilience for, freight transportation.

Design/methodology/approach

The study utilizes semi-structured interviews, supplemented by public company documents, from four actors in one transport supply chain, one transport buyer, one transport coordinator, and two transport providers of different sizes.

Findings

The scope and perceived importance of the different pressures varied between the actors. The transport buyer presented a partial understanding of pressures in transportation, perceiving pressures outside the industry as well. The actors experienced different pressures related to reduced climate impact and improved climate resilience, respectively, and there were differences regarding which pressures triggered the implementation of measures both between actors, and between reducing climate impact and improving climate resilience. Additionally, some pressures spread throughout the chain.

Research limitations/implications

The study advances the understanding of how institutional pressures affect freight transport system actors differently, with consideration for moderating factors such as size, country, and degree of internationalization.

Originality/value

The study's originality lies in its transport chain perspective, its consideration to experienced pressures by both strategic and operational levels within the companies, and integration of perspectives on climate impact and climate resilience.

Paper 22 - Full Paper

Designing pre-positioning systems for military equipment under conditions of uncertainty

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Keywords: Defence preparedness; Military stockpiling; Strategic pre-position stocks; Military supply chain resilience

Purpose

The effective availability and employment of military equipment and essential supplies depend on systems of pre-positioned stocks that enable timely access to resources across branches of the armed forces and ensure interoperability with NATO forces. The paper addresses the concept of pre-positioning and analyses how such systems can be designed to sustain combat capability over time under conditions of uncertainty.

Design/methodology/approach

The study adopts a qualitative case study approach informed by defence logistics, inventory management, and resilience theory. Empirical data are collected through semi-structured interviews with officers in the Swedish Armed Forces and analysed using thematic analysis.

Findings

The findings identify three categories of stocks, standby, mobilisation, and pre-positioned, and specify the types of resources appropriate for each category. The study identifies key drivers, enablers, and risks shaping pre-positioning strategies, resulting in geographic dispersion, accessibility, supporting units, governance arrangements, and defence industry capacity replenishment. Persistent tensions are observed between efficiency-oriented logistics principles and resilience-based requirements.

Research limitations

The study is limited to a single national context and a qualitative dataset. Nevertheless, the findings are considered transferable to countries with defence structures, strategic conditions, and geographic characteristics like those of Sweden.

Practical implications

The findings enhance understanding of defence logistics planning, particularly with respect to vulnerabilities, trade-offs, and decision-making in pre-positioning strategies.

Originality

The study offers novel empirical insights into pre-positioning in a contemporary security environment characterised by a threat landscape distinct from that of the Cold War.

Paper 23 - Full Paper

Techno-economic assessment of renewable energy fuels in construction logistics

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Keywords: Sustainability; Construction logistics; Electrification

Purpose

The paper investigates the techno-economic feasibility of replacing diesel with renewable energy in construction logistics. Construction transport is emission-intensive and slow to adopt new technologies; therefore, understanding cost structures and sustainability gains is critical. The paper quantifies how logistics costs are distributed and compares performance across energy sources.

Design/methodology/approach

A techno-economic analysis is applied using operational data from construction transport and scenario modelling. Costs include salary, vehicle hours, idling, maintenance, and opportunity costs. Diesel, hydrogen, and battery-electric pathways are evaluated.

Findings

While diesel provides low operating costs, renewable alternatives can become competitive considering reduced idle energy use and long-term maintenance, and higher drivetrain efficiency. Hydrogen is efficient with heavy mid- and long-range transportation (or when machines cannot be in electric grid), but non-existing hydrogen markets limit economic benefits, whereas electricity demonstrate strong potential in repetitive, short-distance logistics cycles (or when machines are in grid – such as cranes). Research limitations/implications (if applicable) The analysis is based on representative but simplified transport profiles and evolving energy prices. Practical implications (if applicable) The findings support construction actors in evaluating transition pathways, highlighting where renewable fuels can reduce operating costs and emissions, and where infrastructure investments are critical. Societal implications (if applicable) Adopting renewable fuels can reduce emissions from construction, supporting climate targets and improving urban air quality.

Originality/value

It provides a holistic cost breakdowns of renewable-energy construction logistics that offers actionable insights for industry stakeholders.

Paper 24 - Full Paper

From priorities to acquisition: defense materiel acquisition in six nato countries after 2022

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Keywords: Defense materiel acquisition; Defence procurement governance; Procurement policy–implementation gap; NATO Europe defense capability coordination

Purpose

Following Russia's invasion of Ukraine in 2022, European allies have announced rapid rearmament, enhanced interoperability, and stronger coordination through NATO and the EU. This study examines how these ambitions are reflected in practice by analysing the relationship between stated policy priorities and actual defence materiel acquisition in six European NATO countries: Finland, France, Germany, the Netherlands, Poland, and the UK.

Design/methodology/approach

The study employs a comparative design combining qualitative content analysis of strategic and policy documents published 2022-2025 with mapping of procurement outcomes based on publicly available data on defence budgets and major system acquisitions.

Findings

Despite a largely shared legal framework, states pursue divergent procurement practices and strategic objectives. Policy documents vary in structure, terminology, and specificity, limiting cross-national comparability and alliance coordination. Procurement outcomes show limited short-term adaptation at the major system level, reflecting long acquisition cycles, and largely follow established national trajectories, including continued investment in legacy systems. States emphasise national control over critical technologies and industrial capacity while promoting interoperability, creating persistent governance tensions. Supreme audit institutions report weak linkages between stated priorities and delivery.

Research implications

The study advances defence acquisition research by systematically linking policy priorities to observable procurement outcomes and highlighting the constraining role of institutional inertia.

Practical implications

The findings underscore the need for clearer prioritisation, stronger translation of strategic intent into acquisition practice, and improved transparency to support alliance coordination. Original This paper provides a rare comparative assessment of post-2022 defence materiel acquisition across multiple European NATO countries.

Paper 25 - Full Paper

When procurement meets innovation: institutional constraints on dual-use technologies in Swedish defense

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Keywords: Dual-use innovation; Defence procurement and start-ups; Civilian-defence incubators; Defence organizational inertia

Purpose

There are strong political ambitions to integrate civilian start-ups and dual-use technologies into defence capability development. In practice, start-ups face difficulties accessing military end-users, while the Swedish Armed Forces (SwAF) struggle to absorb and operationalise new technologies. This study examines why these misalignments persist, focusing on defence procurement procedures, institutional barriers, and innovation incubators.

Design/methodology/approach

The study is based on qualitative data from semi-structured interviews with SwAF officers and representatives from civilian and defence incubators, complemented by participation in relevant conferences. The empirical was collected between 2014 and 2016 and analysed using thematic coding.

Findings

Despite political ambitions to integrate civilian start-ups, the Swedish defence innovation system remains dominated by incumbent actors and exhibits limited absorption of start-up innovation. Procurement procedures designed to ensure reliability, regulatory compliance, and lifecycle support constrain early-stage engagement with small firms. Civilian incubators, optimised for open markets, rapid scaling, and venture financing, are poorly aligned with defence requirements related to security, secrecy, and long acquisition cycles. In contrast, defence-oriented incubators, such as NATO's DIANA, are better adapted to military innovation needs.

Research implications

The study contributes to defence innovation research by showing how differing incubation logics shape innovation trajectories and advancing innovation intermediation in security-sensitive contexts.

Practical implications

The findings inform policymakers on how intermediary structures and procurement models can improve start-up accessibility while safeguarding operational readiness and technological sovereignty. Original This paper provides a unique empirically grounded comparison of civilian and defence-oriented innovation intermediaries, demonstrating how institutional inertia constrains adoption of dual-use technologies.

Paper 26 - Full Paper

Humanitarian logistics at sea: maritime transport, conflict, and geopolitical vulnerability

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Keywords: Humanitarian logistics; Maritime transport; Armed conflict; Geopolitical vulnerability; Transport chokepoints

Purpose

Humanitarian logistics research has focused on ground and air transport, with limited discussion of maritime transport. This paper frames humanitarian logistics within the wider maritime transport system, highlighting how geopolitical vulnerability, conflict, and chokepoints shape humanitarian supply chains at sea.

Design/methodology/approach

Three case studies are examined: Ukraine, focusing on maritime transport as an outward flow of humanitarian supplies from a conflict zone; Gaza, examining attempts to deliver humanitarian aid into a besieged territory; and Yemen, analysing humanitarian maritime operations at a critical geopolitical chokepoint. The analysis adopts a systems-oriented perspective, situating humanitarian logistics within global maritime networks and security dynamics.

Findings

The cases reveal distinct but interconnected humanitarian maritime functions: enabling global food and relief supply chains, attempting access to blockaded populations, and sustaining humanitarian lifelines through strategically vulnerable chokepoints. Across cases, maritime transport is both an enabler and a site of political, security, and operational risk. Research limitations/implications (if applicable) Future research should integrate primary data and further theorise humanitarian–commercial maritime interdependencies. Practical implications (if applicable) Findings highlight the need for closer alignment between humanitarian actors, maritime authorities, and security stakeholders in conflict settings. Societal implications (if applicable) The paper informs humanitarian policy debates on access, neutrality, and the protection of maritime corridors in armed conflict.

Originality/value

The paper advances humanitarian logistics research by embedding it within maritime transport systems and geopolitical analysis.

Paper 28 - Work in Progress

Managing change in cross-disciplinary research projects in times of crisis

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Keywords: Planned change; Cross-disciplinary; Unplanned change; Multidisciplinary project; Lewin's change model

Purpose

Academic research projects, particularly those within cross-disciplinary domains, must establish and sustain adaptive and resilient environments to ensure the delivery of high-quality research outcomes. This requires navigating disciplinary complexity, managing environmental uncertainty, and addressing planned and unanticipated changes. This paper details the change experiences and management outcomes of a 6-year (2019-2025) cross-disciplinary project within the defense sector in Sweden. The purpose of this study was to highlight important, often ignored, aspects of change involved within multidisciplinary, cross-year academic projects.

Design/methodology/approach

Drawing on experiential insights from principal investigators, work package leaders, and project partners, this paper examines the project's responses to six major world events including the 2019 Pandemic (CoVid-19), the MV Ever Given blockade of the Suez Canal, the Russian invasion of Ukraine, ongoing Houthis conflict in the Red Sea, Swedish NATO membership, and the anticipated increase of NATO defense spending from 3% to 5% of GDP. The study highlights their impacts on project dynamics, challenges, and adaptive strategies undertaken.

Findings

The analysis supports the applicability of Lewin's Change model to better understand the diverse contexts within which multidisciplinary academic projects operate. However, given the continuous, contextual, and interactive nature of the project, the study results indicate a potential for further development of Lewin's model to address the complexities of change management that may occur within multidisciplinary projects.

Originality

This study tests and extends a long-standing model – Lewin's change model – within the context of unplanned change in multidisciplinary academic research.

Paper 30 - Work in Progress

Assessing digital adoption readiness in complex, project-oriented supply chains: A literature-based conceptual perspective

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Keywords: Digital adoption readiness; Pre-adoption stage; Complex Supply chain

Purpose

The paper aims to clarify how digital adoption readiness is conceptualized in existing research and to assess its relevance for organizations operating in complex, project-oriented supply chains. This clarification supports a more coherent understanding of readiness considerations at the pre-adoption stage of digital adoption.

Design/methodology/approach

The study adopts a a structured literature review approach to examine research on digital adoption and organizational readiness in supply chain and organizational contexts. Peer-reviewed journal articles are systematically identified and analyzed to extract key readiness dimensions and theoretical perspectives relevant to the pre-adoption stage. The review covers digital transformation, technology adoption, and readiness assessment in complex and project-oriented supply chains, forming the basis for subsequent conceptual framework development.

Findings

The paper identifies recurring readiness dimensions and conceptual patterns in the literature on digital adoption. Preliminary insights from the literature indicate fragmentation across existing readiness models and limited consideration of project-oriented supply chain contexts.

Research limitations

As a work-in-progress, this paper is limited to a literature-based synthesis and conceptual development. Further research will empirically examine and validate the proposed readiness dimensions across organizational contexts.

Originality/value

The paper initiates a structured synthesis of literature on digital adoption and organizational readiness, focusing on the pre-adoption stage in complex, project-oriented supply chains. It offers early value by organizing fragmented knowledge for researchers and supporting informed early-stage digital adoption discussions for practitioners.

Paper 32 - Work in Progress

Resilience of critical societal functions : a supply chain perspective

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Keywords: Societal functions; Resilience; Supply chain; Sustainability; Preparedness

Purpose

Critical societal functions, which meet citizens' basic needs and ensure social order are entwined in global supply chains through social, material, and institutional processes. Therefore, supply chain resilience, which relies on adaptability, preparation, and agility to overcome shocks is needed to ensure continuity of these functions. This paper explores the supply chain factors contributing to critical societal functions to establish a societal resilience supply chain framework.

Design/methodology/approach

Data is collected through workshops, focus groups, and interviews with relevant stakeholders in Finland. Critical functions are mapped and discussed via future pathway exercises and causal loop diagrams.

Findings

From a societal perspective, critical functions are determined through system, institutional, and operational levels, ensuring that basic needs are met, responsibility among actors is established, and continuity is operationalised. The complex fabric of society brings tensions into the functions' supply chains. The framework developed includes equality and sustainability as additional factors, which diversifies the stakeholder profile and mechanisms for adaptability to societal changes/shocks. Research limitations/implications (if applicable) Findings have implication for resilience policy and practice in Finland, and can be expanded to include e.g. the Nordic context.

Originality/value

Considering supply chain factors for critical functions highlights practical strengths and weaknesses and provides insights to realign functions and stakeholders to emphasise equality and sustainable development.

Paper 33 - Work in Progress

The implications of nato membership for critical infrastructure development in Finland and Sweden

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Keywords: Critical infrastructure; Critical event analysis; NATO; Finland; Sweden

Purpose

Recent crises and wars have shown the need for security of supply, resilient supply chains and societal resilience. The changing geopolitical landscape following international tensions and the Russian invasion of Ukraine has resulted in Finland and Sweden becoming NATO members, emphasizing a change in the approach to critical infrastructure development. The purpose of this paper is to understand the needs to develop critical infrastructure in Finland and Sweden.

Design/methodology/approach

The paper analyses critical events, decisions and patterns that support, or alter the prioritisation of critical infrastructure development. Specific critical events include the cutting of undersea cables, heavy rains, the Covid-19 pandemic and electrical disturbances. Critical event analysis is employed to analyse these events, providing a parsimonious explanation of a puzzling outcome.

Findings

The findings will contribute to knowledge on the infrastructure development in Finland and Sweden where the analysis highlight crisis preparedness and prioritisation mechanisms in the light of effective support of societal resilience.

Research limitations/implications

The paper study several critical events and the outcome of these. Selection of other events in other regions might give different results

Practical implications

The paper intends to give recommendations to public authorities on prioritization.

Social implications

The paper includes a further understanding of Nordic integration not just across defence but crisis preparedness and civil protection.

Originality/value

The paper shows the importance of robust and resilient infrastructure.

Paper 34 - Full Paper

The impact of online grocery service attributes on retailer preference and consumer satisfaction

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Keywords: Online Grocery Retailing; Fulfillment Services; Consumer Behavior; Logistics Plackett–Luce Model; Multinomial Logit Model

Purpose

As online grocery retailing rapidly expands, the impact of fulfillment services on consumer choices and post-purchase satisfaction with online grocery retailers remains uncertain. This study seeks to 1) identify the key fulfillment factors that influence consumers' choice of online grocery retailers and 2) examine the fundamental elements that impact satisfaction evaluations after the consumer shopping experience.

Design/methodology/approach

Utilizing survey data from Korean online grocery consumers, this study employs the Plackett–Luce model to pinpoint fulfillment attributes that influence retailer choice and a Multinomial Logit model to analyze post-purchase satisfaction. By comparing these two stages, the study assesses the asymmetric effects of fulfilment services in online grocery retailing.

Findings

The results indicate that consumers consistently prefer certain fulfillment options, such as fast delivery and eco-friendly packaging, across different retailers. However, their satisfaction evaluations are influenced by a wider range of factors, highlighting a disconnect between what drives the expectation of customers and what influences satisfaction. Practical implications (if applicable) The findings indicate that online grocery retailers should tailor fulfillment roles to different decision stages and adjust their fulfillment strategies and logistics configurations accordingly.

Originality/value

This study, grounded in the Logistics Service Quality Model, incorporates a consumer decision-stage perspective and expands the research framework of fulfillment services in online grocery retailing.

Paper 35 - Full Paper

Understanding resource dependences in electrified freight transport

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Keywords: Electrification; Road freight; Green logistics; Resources; Resource dependency; Case study

Purpose

This research aims to enhance understanding of the interdependencies between actors and their resources to overcome barriers in the transition to electrified freight transportation, a key step towards sustainable logistics systems.

Design/methodology/approach

A case study, based on interviews, was conducted with ten companies actively engaged in the electrification of freight transport systems. A resource dependency perspective was applied to analyse the resources and interdependencies among the actors involved.

Findings

Beyond physical resources such as electric trucks and chargers, the findings emphasise that competence, system understanding, and data are critical resources that are dispersed across multiple actors. Consequently, knowledge-based, technical, financial and operational dependencies were common among involved actors.

Research limitations/implications

The results shed light on the critical role of resources and how decisions regarding which resources to retain internally and which to source from other actors create inherent dependencies, which, in turn, lead to more complex and interconnected systems.

Practical implications

The study highlights the importance for practitioners of determining which resources to own and understanding the dependencies that arise when resources are not owned. The results also indicate which types of resources lead to the most common dependencies.

Originality/value

This study advances resource-based perspectives on sustainability transitions by mapping resource types and holders in freight electrification, offering insights into how physical, financial, and knowledge resources interact across actor groups.

Paper 38 - Full Paper

The role of consumers as suppliers in circular procurement: insights from the consumer electronics sector

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Keywords: Circular procurement; Consumer electronics; Circular economy; Consumers-as-suppliers

Purpose

This study explores how consumers may function as suppliers in circular procurement for consumer electronics. As circular business models rely on returned devices for repair, refurbishment, and resource recovery, understanding consumer willingness to supply end-of-use products becomes essential for procurement.

Design/methodology/approach

The study draws on three focus groups to examine consumers' willingness, ability, and perceived risks associated with returning smartphones, laptops, tablets, and other small electronics. The focus groups were conducted to develop a survey to be administered in spring 2026.

Findings

Contributions include insights into how consumers perceive their potential role as suppliers and how behavioural, psychological, and contextual factors influence their decision to return devices.

Practical implications

The study aims to inform procurement strategies for designing and managing consumer-dependent reverse supply channels. Insights about trust, incentives, convenience, and perceived risks may support organisations in building more reliable and legitimate return systems

Social implications

Understanding consumer-driven inflows can enhance the effectiveness of circular electronics systems, thereby reducing resource extraction, lowering e-waste, and promoting more sustainable consumption patterns.

Originality

The study extends procurement theory by positioning consumers as a non-traditional but essential part of the supply base in circular electronics. It highlights the behavioural dynamics shaping circular sourcing and provides value to scholars and practitioners seeking to operationalise consumer participation in circular procurement.

Paper 40 - Work in Progress

Orchestrating the electric freight landscape: how industry archetypes shape charge point operator practices

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Keywords: Charging Point Operators; Ecosystem orchestration; Sustainability transitions; BEV; Capability

Purpose

Electrification of heavy freight transport constitutes a complex systemic transformation, requiring the integration of traditionally separate logistics and energy systems. This paper examines how the originating sector of a Charge Point Operator (CPO) influences its inherent orchestration capabilities, shaping the practices employed to align ecosystem actors.

Design/methodology/approach

With a point of departure in literature on orchestration capabilities, a deductive qualitative case study approach was used with eight interviews to capture empirical data. The respondents represent actors from both the energy sector and logistics sector. The analytical lens is based on orchestration practices leading to microfoundation of capabilities.

Findings

Preliminary results reveal two distinct CPO capability archetypes: energy-led and logistics-led. Energy-led CPOs adopt grid-centric configurations, focusing on grid access, capacity planning and regulatory compliance, but lack logistics planning capability. Logistics-led CPOs prioritise infrastructure based on delivery routes and depot optimisation, excelling in freight flow control but lacking energy market knowledge.

Research limitations/implications

The findings imply that ecosystem orchestration strategies must be tailored to industry-specific capability profiles. This study shows originality by revealing how industry-specific capability profiles shape ecosystem orchestration during sustainability transitions. Future research should explore how these archetypes evolve under collaborative governance models.

Practical implications

The results provide actionable guidance for firms operating at the intersection of energy and logistics. By identifying capability gaps, organisations can develop targeted collaboration models and contractual mechanisms to ensure utilisation guarantees.

Originality/value

It offers novel insights into collaborative strategies that enable large-scale freight electrification and support the shift towards fossil-free transport.

Paper 41 - Full Paper

Emission scope heterogeneity and climate governance: the role of supplier and customer engagement in climate-related incentive adoption

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Keywords: Scope 3 emissions; Supplier engagement; Customer engagement; Management incentives

Firms are increasingly expected to manage greenhouse gas emissions from both internal operations (Scope 1 and Scope 2) and value-chain activities (Scope 3). Prior research has largely examined supplier engagement, customer engagement, and management incentives as separate levers of climate governance, leaving limited evidence on how engagement practices connect to incentive strategies when firms' emissions exposure differs by scope. To address this gap, this study examines whether firms with higher scope-specific emissions adopt different engagement strategies with suppliers and customers, and whether these engagement choices are associated with climate-related incentives in the short term and the long term.

Using firm-level responses to the CDP Climate Change questionnaire, we examine associations between Scope 1–2 and Scope 3 emissions and CDP-based measures of customer engagement, supplier engagement, and climate-related incentive planning. The findings indicate systematic differences by emission scope. Scope 3 emissions are positively associated with supplier engagement, whereas Scope 1–2 emissions are more closely associated with customer engagement. Supplier engagement is also positively associated with long-term climate-related incentives, suggesting a stronger role for longer-term incentive design in governing value-chain decarbonization. In comparison, customer engagement shows a weaker association with incentive planning, and short-term incentives are only weakly related to engagement practices.

Overall, the study advances climate governance research by empirically linking scope-specific emissions exposure to engagement practices and incentive planning, with implications for decarbonization beyond firm boundaries.

Paper 43 - Full Paper

Time-to-event analysis in spare parts demand forecasting: a systematic literature review

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Keywords: Spare parts management; Survival analysis; Intermittent demand; Maintenance analytics; Aftermarket logistics

Spare parts demand forecasting is a fundamental pillar of data-driven smart maintenance, directly influencing asset availability, service levels, customer satisfaction and overall business performance. However, this task remains notoriously difficult due to the intermittent nature of demand data, which is often complicated by a limitation of historical records. Conventional time-series forecasting techniques frequently fail to adequately capture these complex patterns, necessitating the adoption of more sophisticated methodologies. One particularly promising enhancement involves the application of time-to-event (TTE) analysis—also known as reliability or survival analysis—which leverages data on machine breakdowns and component life cycles to predict future requirements. Despite the potential of these techniques, research on TTE-based approaches for spare part demand forecasting remains highly fragmented. To the best of our knowledge, no systematic review currently exists that focuses specifically on this topic. This paper addresses this gap by conducting a comprehensive systematic literature review of 88 peer-reviewed publications. Furthermore, we identify critical research gaps, in the field of TTE and time series forecasting for spare parts management. Our findings offer valuable insights for both researchers and practitioners operating at the intersection of maintenance and forecasting, ultimately facilitating more proactive and cost-effective inventory strategies.

Paper 44 - Full Paper

Strategic inventory management decision framework for cto manufacturing

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Keywords: Configure-to-order; Make-to-order; Inventory decision framework; Demand variability

This study advances inventory management research in Configure-to-Order (CTO) manufacturing by dveloping and empirically grounding a strategic decision framework that explicitly accounts for demand heterogeneity, lead-time asymmetry, and cost–service trade-offs. CTO environments challenge conventional inventory theories due to high product variety, intermittent demand patterns, and frequent misalignment between planning parameters and actual consumption behavior. Existing approaches often rely on standardized ABC or service-level rules, which inadequately capture this operational complexity. This paper addresses this gap by proposing a demand-segmented, decision-oriented framework that links inventory control logic to observable SKU-level characteristics. The research adopts a rigorous mixed-methods case study design. Qualitative data were collected through structured interviews across core supply chain functions and analyzed using a Supply Chain Management Framework to identify systemic misalignments in demand management, inventory control, and warehouse operations. Quantitative analysis draws on two years of transactional ERP data and applies demand profiling, variability metrics, inventory aging analysis, and scenario-based simulations to evaluate alternative inventory policies. Demand is segmented at the SKU level using volume, variability, and lead-time dimensions, enabling systematic comparison of strategic inventory options. The results demonstrate that uniform inventory policies systematically inflate stock levels and obscure improvement potential in CTO manufacturing. The proposed framework prescribes differentiated strategies, including Kanban replenishment, statistical forecasting, selective buffering, and buy-to-order, aligned with demand segment characteristics and economic impact. Empirical findings indicate substantial potential for reducing inventory value, releasing working capital, and improving service performance without increasing operational risk. This contribution strengthens theory and relevance.

Paper 45 - Work in Progress

Humanoid robots in logistics operations - a literature review

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Keywords: Humanoid robots; Automation; Supply chain; Logistics

Purpose

This study aims to map how humanoid robots are expected to be incorporated into supply chains and logistics operations, for example in packaging tasks and loading bay activities.

Design/methodology/approach

The paper is a literature review examining both grey and academic literature.

Findings

The commercialization of humanoid robots is expected to accelerate in the coming years, and their ability to perform a range of functions in supply-chain settings has already been demonstrated in publicly shared tests. However, as the technology continues to develop, uncertainty remains regarding the roles humanoid robots will ultimately assume in supply chains. This paper examines how humanoid robots are expected to be commercialized and identifies key conditions, functions, and locations where they are likely to appear in supply-chain and logistics operations. It also investigates anticipated operational effects and potential impacts on sustainability parameters.

Research limitations/implications

The study is limited to literature and reports written in English. The paper thus does not explore developments in the PRC, Japan, South Korea and elsewhere which is not reported in English.

Practical implications

The paper categorises and describes use cases of humanoid robots in logistics operations.

Social implications

The results will help researchers and stakeholders to estimate potential benefits and drawbacks of automations through the use of humanoid robots.

Originality/value

The paper explores the quickly evolving area of humanoid automation.

Paper 47 - Full Paper

Inventory classification to improve inventory control policies for wholesale grocery co-operatives

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Keywords: Inventory classification; Inventory control policy; Groceries; Wholesale co-operative

Situated between volatile supplier conditions and highly fragmented retail demand, grocery wholesalers face greater challenges of inventory management than individual stores. Furthermore, wholesale assortments can reach tens of thousands of SKUs, making inventory replenishment decision by item extremely challenging. This study aims to provide a systematic framework to classify wholesale grocery products based on their demand characteristics and to suggest differentiated inventory control policy by each inventory profile. This study applies a rule-based analytical approach to classify grocery products based on their demand characteristics including perishability, seasonality, sales volume, and sales frequency. It further proposes replenishment policies (optimal inventory levels, replenishment cycles, order quantities, etc.) considering sales volume and sales volatility by product group. These customized replenishment policies per product group are expected to minimize excess inventory and stockouts while improving inventory turnover and order fulfillment rates. For products with low sales volatility where demand forecasting is straightforward, establishing automated ordering and regular delivery systems is also expected.

Paper 48 - Full Paper

Individual-level coping and supply chain resilience under catastrophic earthquakes

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Keywords: Supply chain resilience; Coping theory; Earthquake

Purpose

This study examines how individuals in supply chains (SC) psychologically cope with earthquakes and how these coping mechanisms impact organizational supply chain resilience (SCR). This paper aims to bridge the gap between understanding individual-level responses and organizational SCR after a natural disaster.

Design/methodology/approach

Data were collected from 40 semi-structured interviews in Türkiye's regions affected by the 6 February 2023 earthquakes, across the automotive, textile, and food sectors. Informants covered top management and operational roles. In addition, relevant official reports published on the earthquakes and their impacts were reviewed to triangulate and contextualize the interviews. Interview data were analysed using coping theory at the individual level as emotion-focused and problem-focused coping responses, followed by firm-level aggregation to derive organizational practices and map them to SCR stages. We included SCR stages as response, recovery, learning in our research frame.

Findings

Findings indicate that individual coping can influence the SCR. After the earthquake, emotion-focused coping (e.g., avoidance, withdrawal) reduced stress but delayed communication, coordination, and decisions, weakening response and slowing organizational recovery. Problem-focused coping (e.g., redesigning routines, reallocating tasks) was less common but supported organizational continuity. At the organizational level, firms introduced practices such as immediate protective actions, psychological support mechanisms, efforts to restore disrupted supply chain activities, and post-event learning initiatives. Original This research links the individual coping mechanisms to SCR, as both organizational and individual levels influence SCR.

Paper 49 - Full Paper

The adaptive cycles of regenerative supply chains

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Keywords: Regenerative supply chain; Adaptive cycles; Panarchy Theory; Circular bioeconomy; Socio-ecological resilience

Purpose

Bio-based materials and products are being developed to support the regenerative nature of supply chains beyond the technical aspects of the circular economy. The purpose of this paper is to extend an understanding of regenerative supply chains (RSC) through the adaptive cycles framework of panarchy theory. The phases of the adaptive cycle help explain the qualities of regenerative dynamics embedded in RSCs, and how both technical and biological cycles contribute to these qualities and enhance socio-ecological resilience.

Design/methodology/approach

The research applies adaptive cycles to elucidate the regenerative dynamics of supply chains, connecting the phases of technical and biological cycles to the qualities of the RSC. An interview study with respondents from humanitarian organizations supports the mapping of adaptive cycles to RSCs.

Findings

Adaptive cycles allow us to expand our view of RSC beyond regenerative processes: As socio-ecological systems, RSCs can operationalize a circular bioeconomy and integrate biological cycles into the 10R framework of the circular economy. At the same time, the paper highlights instances in which bio-based products that may be regenerative are developed in a throwaway fashion, thereby undermining socio-ecological resilience. Implications: The research offers an interesting value-addition by elucidating the core qualities of regenerative dynamics in the supply chain. It also helps practitioners understand RSC and apply its core qualities in their quest to make the supply chain regenerative.

Originality/value

This research is the first step toward understanding RSC through the adaptive cycle and offers commentary on its feasibility for advancing the circular bioeconomy.

Paper 50 - Full Paper

Dynamic capabilities for designing and operating returnable transport packaging systems

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Keywords: Circular economy; Dynamic capabilities; Organizational capabilities; Reusable packaging systems; Sustainable packaging

Purpose

This study aims to identify and categorize dynamic capabilities through which firms mobilize and reconfigure resources and operational processes to design and operate sustainable returnable transport packaging systems. It further distinguishes these capabilities according to their relative criticality and required flexibility.

Design/methodology/approach

A qualitative study drawing on interviews and archival data from the automobile and grocery retail industries, using dynamic capabilities theory as an analytic lens to examine the design and operation of returnable transport packaging systems.

Findings

We identify three interrelated sets of dynamic capabilities: for sensing and articulating the needs of returnable transport packaging systems, for translating these needs into the design and operation of these systems, and for reconfiguring underlying resources and processes to ensure the evolutionary fitness of these systems.

Research implications

This study extends the literature on returnable transport packaging systems by showing that their design and operation depend on a distinct set of dynamic capabilities rather than on isolated operational practices. Specifically, it demonstrates that these capabilities differ in criticality and required flexibility, shaping how firms build, adapt, and sustain returnable transport packaging systems over time.

Practical implications

Managers get practical guidance by clarifying which organizational capabilities are needed to design and operate returnable transport packaging systems, which are most critical, and how much flexibility each capability requires.

Originality/value

Returnable transport packaging systems are presented as capability-dependent systems by identifying the resources and processes required for their design and operation, and by distinguishing the degree of flexibility required for different capabilities over time.

Paper 52 - Full Paper

Beyond accuracy: nesting forecasting and forecastability in supply chain demand forecasting

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Keywords: Demand forecasting; Artificial intelligence; Supply chain management; Strategy-as-practice; Sociomateriality; Delphi study

Purpose

This research investigates how AI-enhanced demand forecasts gain strategic significance in supply chain management. Rather than treating forecasts as more or less accurate inputs, the study examines the processes through which demand forecasts become acted upon and made consequential.

Design/methodology/approach

The study adopts a strategy-as-practice and sociomaterial perspective and reports findings from a Delphi study with industry experts and supply chain consultants.

Findings

In the current literature, AI-enhanced forecasting is predominantly legitimized over traditional methods through metric-based evaluations against realized demand, while implementation challenges are largely treated as barriers to be overcome. Our findings indicate that forecasts gain strategic significance through complex organizational practices and adjustment mechanisms that respond to deviation between forecasts and realized values. As a result, forecasts in practice function not as neutral predictive tools but as mechanisms that create commitments, opportunities, and risks.

Research limitations/implications

The study relies on expert judgments rather than direct observation of practices within specific firms. Future research could combine this practice-based perspective with empirical studies across different supply chain contexts.

Research implications

The paper reframes forecasting accuracy as an emergent accomplishment of supply chain arrangements and decision-making rather than a neutral input to planning processes.

Practical implications

Managers should treat AI-enhanced forecasts as scenarios forming commitments and opportunities, rather than as neutral planning inputs.

Originality/value

The paper offers a novel sociomaterial reframing of AI-based demand forecasting that explains how forecasting numbers come to count strategically in practice.

Paper 54 - Full Paper

Digital and sustainable supply chain: a social capital perspective

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Keywords: Digital Transformation; Supply Chain Management; Sustainable Performance

This research examines the mediating role of social capital in the relationship between digital transformation (DT) and sustainable performance across developed and developing economies. It aims to uncover how economic context shapes these mediating pathways. Data was collected through structured surveys using questionnaires from 150 respondents, which were collected from LinkedIn and international trade fairs. Data were analysed using structural equation modelling in SmartPLS 4. Digital transformation in supply chains is a key driver of sustainability, boosting economic efficiency, social equity, and environmental performance. However, technology alone is not enough. The role of social capital—whether cognitive, relational, or structural—acts as a force multiplier, strengthening the link between digital tools and sustainable outcomes. This study further develops social capital theory by demonstrating how the level of economic development influences social capital and identifying the aspects that facilitate the connection between digital transformation and sustainability. It questions one-size-fits-all mediation models, showing that relational capital does not always mediate performance, even when influenced by DT. However, the roles of structural and cognitive capital differ greatly depending on the context. This emphasises the importance of developing strategies that are customised for each unique context, focusing on the social capital elements most pertinent to the local situation to achieve sustainable outcomes.

Paper 55 - Work in Progress

Resilience of grocery distribution networks: a case of warehouse disruptions in Sweden.

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Keywords: Food supply chains; Transport disruption; Resilience; Sweden

Global food supply chains are increasingly designed for efficiency through just-in-time practices, low inventories, and tightly scheduled transport operations, but these features also amplify vulnerability to transport disruptions, particularly in import-dependent countries. This paper examines transport disruptions from the perspective of food retailers and analyses how failures at key transport nodes/links affect national food distribution patterns and regional supply vulnerability. Using Sweden as a case study, the study analyses how disruptions propagate through the food transport network and generate regional impacts. The study applies a simulation-based empirical approach using detailed shipment data from Swedish food retailers, including origins and destinations, transport modes, shipment weights, and delivery frequencies. The paper contributes to theory by advancing a transport-focused, spatially explicit perspective on food supply chain resilience. Managerially, the results support retailers and logistics providers in identifying critical transport dependencies and designing more robust distribution strategies. Societally, the findings inform policy discussions on safeguarding food security and ensuring equitable regional access to food during disruptive events.

Paper 56 - Full Paper

Powerful relationships: accelerating the adoption of heavy-duty battery electric trucks

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Keywords: Logistics service provider; Shipper; Green logistics; Sustainable road freight; Power; Case study

Purpose

Advancing sustainable road freight through the adoption of heavy-duty battery electric trucks (HBETs) requires an understanding of how power dynamics between logistics service providers (LSPs) and shippers can either enable or hinder the transition. Therefore, the purpose is to explore how power in LSP-shipper relationships affects the adoption of HBETs in logistics networks.

Design/methodology/approach

The study is based on a single-case study with four embedded cases. Data were collected through interviews with one LSP and four shippers engaged in electrification initiatives. A power-based analytical lens is applied to examine how different power bases influence the adoption of HBETs.

Findings

The study shows that existing power asymmetries allow shippers to use coercive and reward power to push for electrification. Simultaneously, implementing HBETs gives the LSP operational experience that strengthens its expert and referent power, enabling it to attract new shippers and accelerate adoption across its network.

Research limitations/implications

The study contributes to green logistics research by illustrating how power in LSP-shipper relationships shapes HBET adoption, with particular attention to the development of electric road freight.

Practical implications

LSPs and shippers can use these insights to understand how power dynamics may support HBET implementation and foster sustainable transport. In particular, LSPs can gain knowledge about how different forms of power can be leveraged to advance adoption.

Originality/value

Contrary to previous research, this study adds knowledge about how power in LSP–shipper relationships can drive HBET adoption in logistics networks, thus accelerating the transition to sustainable road freight.

Paper 57 - Work in Progress

Localizing food supply chains: insights from wholesalers, retailers, and platforms

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Keywords: Local food supply chain; Wholesaler; Retailer; Digital Platform; Critical Success Factors

Purpose

To mitigate vulnerability to geopolitical disruptions and sustainability-related issues, increasing attention has been directed towards the localization of food supply chains. To expand the operations of local food supply chains (LFSC), the active participation of intermediaries is crucial. Hence, we aim to identify critical success factors (CSFs) facilitating the participation of wholesalers, retailers, and digital platforms in LFSC.

Design/methodology/approach

We have adopted a multiple case study approach and conducted 20 semi-structured interviews across 8 theoretically sampled cases, with durations from 26 to 90 minutes.

Findings

Our analysis reveals that CSFs span micro-, meso-, and macro- levels. Macro-level factors like "customer awareness", "economic condition, "socio-cultural condition", and "governmental support" establish market and institutional context for intermediaries. Whereas the meso-level factors, such as "customer demand", "adaptation in practice", and "availability of local producers", determine their operational condition. Moreover, micro-level factors such as "marketing mix", "sourcing practice", "logistics design", "empowerment", "engagement", "digitalization" and "service augmentation" enhance such intermediaries' internal ability to participate in LFSC. However, these micro-level factors were apparent to varying extents across the cases.

Research implications

We contribute to the literature by identifying multi-level CSFs enabling the integration of local food into mainstream and digital distribution channels.

Originality/value

Unlike prior studies, this research examines the multiple intermediaries' perspectives on the CSFs for localizing food supply chains and demonstrates that these factors span multiple levels.

Paper 58 - Full Paper

Enabling supply chain transformation - a dynamic capabilities perspective based on the pharma & life sciences industry

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Keywords: Dynamic Capabilities; Micro Foundations; Supply Chain Transformation

Purpose

This research explores how dynamic capabilities and their microfoundations enable the transformation of supply chains in response to long-term structural pressures.

Design/methodology/approach

Theoretically grounded in the dynamic capabilities framework, this study employs a qualitative, explorative research design. Primary data was gathered through 23 semi-structured interviews with senior industry experts in the pharma & life sciences industry.

Findings

Supply chains are forced to continuously transform. Transformational measures include diversification, localization, reshoring, and modularization. These measures are supported by dynamic capabilities and their microfoundations, such as a learning culture, talent development, efficient communication, and adoption of digital technologies. The observed dynamic capabilities are integrated into a capability framework that also highlights barriers and enablers. Research limitations/implications (if applicable) The study develops an empirical framework from qualitative data collected in the pharma & life sciences industry. The findings are expected to be transferable to similarly complex supply chains (e.g., semiconductors), while broader application may require further research and adaptation. Practical implications (if applicable) The results can inform capability-building programs and initiatives aimed at enabling change and enhancing resilience, especially in complex supply chains. Societal implications (if applicable) Strengthening pharma & life sciences supply chains enhances security and resilience, ensuring medicine availability and healthcare stability during disruptions.

Originality/value

This study addresses the scarcity of empirical evidence in dynamic capability literature by navigating the tension between rigid regulation and transformation. It offers critical insights into the enablers and barriers to developing dynamic capabilities within complex supply chains.

Paper 60 - Work in Progress

Disruption as endogenous instability: a second-order mechanistic model

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Keywords: Disruption; Mechanics of Transitions; Mathematical Modelling

Efficiency improvements are widely promoted as a pathway to sustainability in operations and supply chain management, yet they are persistently associated with demand expansion and rebound effects. Recent work has formalised this relationship as a first-order mechanism, showing how efficiency gains interact with system elasticity to drive service expansion under conditions of structural stability. However, this linear formulation cannot explain disruption, which has become a central research topic in operations management and socio-technical transitions research.

Disruption is typically described qualitatively or modelled as an exogenous shock, rarely formalised as an endogenous outcome of internal system dynamics. Hence, the conditions under which efficiency-driven systems endogenously lose stability remain under-specified. This paper addresses this gap by developing a second-order formulation conceptualising disruption as an internal system instability emerging from accumulated efficiency-induced demand expansion. Building on a first-order relationship between efficiency, elasticity, and service growth, the second-order model introduces scaling pressure as an accumulated state variable capturing the intensification of system use and connectedness over time. Disruption happens when accumulated scaling pressure exceeds the system's adaptive capacity, resulting in a loss of structural stability and the transition from linear to nonlinear dynamics.

We formalise this transition through a threshold condition distinguishing stable expansion from endogenous instability. The paper advances a mechanistic explanation of regime breakdown and reconfiguration that complements existing qualitative and simulation-based approaches. The proposed second-order formulation provides a foundation for modelling collapse and recovery in operations and supply networks, opening new avenues for analysing the mechanics of transitions in sustainability research.

Paper 61 - Work in Progress

Trade-offs and enabling implications of structural flexibility: a multiple case study

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Keywords: Structural Flexibility; Performance; Trade-offs; Supply Chain Design

Purpose

Structural flexibility is recognized as an important capability for enhancing resilience and mitigating supply chain risks. Existing research offers limited insight into how different types of structural flexibility translate into specific performance and trade-off outcomes. The purpose of this study is to examine how different types of structural flexibility are linked to performance and trade-off outcomes.

Design/methodology/approach

The study adopts a multiple-case study design, interviewing companies that use practices forming different types of structural flexibility. Through qualitative analysis, the study explores how these structural flexibility types are linked to specific performance and trade-offs outcomes.

Findings

The findings show that different structural flexibility types are associated with distinct performance and trade-off outcomes. While certain types support specific performance goals, they simultaneously cause trade-offs, indicating that structural flexibility is not a universally beneficial capability. Theoretical contributions This study contributes to structural flexibility theory by conceptualizing structural flexibility not only as a capability to achieve, but as a deliberate design choice resulting in certain performance and trade-offs outcomes. Managerial/ Practical implications The results provide managers with guidance on how different structural flexibility types can be aligned with specific performance objectives, while potentially avoiding trade-offs.

Social implications

By understanding the performance and trade-off outcomes of structural flexibility, the study contributes to the development of more resilient and stable supply chains, thereby helping to reduce vulnerability to disruptions and product scarcity.

Paper 63 - Full Paper

Clean energy transition in landlord container seaports: a dynamic capability perspective

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Keywords: Port; Sustainability; Disruptions; Tariffs; Trade-off

This study explores the multifaceted challenges of implementing sustainable solutions at seaports, with a particular focus on the upstream impacts on supply chain and the complexities introduced by global trade dynamics. The purpose of the research is to investigate how modernization and sustainability initiatives at seaports affect not only direct port operations but also the broader supply chain, particularly concerning greenhouse gas (GHG) emissions. The methodology involves case studies of two seaports, utilizing both qualitative and quantitative data to assess various sustainable initiatives. The analysis incorporates stakeholder perspectives and evaluates the contextual variables influencing outcomes, including regional regulations, energy mixes, and operational interfaces between sea and hinterland, under changing global trade dynamics. The study's contribution lies in highlighting the risk of merely shifting emissions upstream rather than achieving holistic reductions, and underscores the need for supply chain-wide perspectives in policy and practice in particular during disruptive events like e.g. sudden change of import/export tariffs. Findings could serve as benchmarks for policymakers, port authorities, and port terminal operators, emphasizing the importance of integrated approaches to sustainability that account for systemic complexities and evolving industry challenges.

Paper 64 - Work in Progress

Crowd humanitarian logistics

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Keywords: Self-organization; Individuals; Response; Humanitarian

The outbreak of full-scale war in Ukraine on 24 February 2022 was marked by the emergence of multiple individual humanitarian logistics initiatives, particularly across Ukraine and through the Polish border. This paper seeks to conceptualize such fragmented humanitarian logistics through a threefold theoretical framework combining insights from social anthropology, crowd logistics, and humanitarian supply chain management. We also aim to explore how to better integrate these practices into humanitarian supply chains, alongside their more traditional actors. Social anthropology values these forms of self-organization, which it clearly contrasts with established humanitarian logistics, criticized for distorting the field and ethos of humanitarian aid through the application of "Fordist principles". We therefore begin by examining the respective perspectives on crowd humanitarian logistics initiatives from the viewpoints of humanitarian SCM and social anthropology, drawing on the existing literature. We then analyse, through semi-structured interviews with all relevant actors (engaged citizens, NGOs, and humanitarian logistics service providers), these spontaneous logistics movements operating within conflict settings. This research enables us to characterize individual humanitarian logistics initiatives in Ukraine by mobilizing the overarching concept of crowd (humanitarian) logistics. It also examines the arguments put forward by field actors, humanitarian practitioners, and academics from the two disciplines involved regarding the observable modalities of crowd humanitarian logistics initiatives, as well as their relevance and respective limitations. By extending the analysis beyond disciplinary boundaries, this contribution enriches the still limited body of research on the role of individuals in humanitarian supply chains.

Paper 66 - Full Paper

Evaluation of suppliers of critical products as preparedness for crisis

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Keywords: Resilience evaluation; Critical product; Supplier selection; Security of supply

Swedish authorities are tasked with securing critical product supply in crisis situations. As response, authorities may sign total defense contracts with industry to secure supply of the products they deem critical. To select companies, evaluation of the production ability, capability and resilience of the companies is important. This study is based on empirics from a supplier evaluation and then comparing it to parameters that are seen as important according to studies in academic literature. In literature both qualitative and quantitative evaluation of production capacity are presented, sometimes using parameters that are difficult to measure or evaluate effect of. In this empirical study an Authority Agency selected five products deemed critical in some scenarios. The institute RISE then identified and contacted 15 companies willing to perform regular production and /or emergency production of such products. The companies were visited, interviewed and a qualitative evaluation was used by five evaluators. In the evaluation of the emergency production capability, a diverse set of parameters were used: Owner structure, economic stability, personnel and competence, certificates and quality performance, production equipment, maintenance and logistics. Afterwards the evaluators were interviewed to find what parameters were seen as useful and which parameters were seen as difficult to evaluate. The evaluation was then compared to findings in literature. The study contributes with empirical experience of evaluating emergency production capability and discussing this in the light of existing theoretical models.

Paper 68 - Work in Progress

Challenges of hydrogen fuel-cell trucks in the nordics: logistics service providers' perspective

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Keywords: Adoption barriers; Emerging technologies; Sustainability transitions

Purpose

This paper examines what kind of challenges hinder the adoption of hydrogen-powered heavy-duty trucks and identifies lessons from early adopters to overcome these barriers.

Design/methodology/approach

The study applies a qualitative research design that draws on two complementary datasets: interviews with Finnish LSPs regarding anticipated challenges related to the adoption of hydrogen trucks, and interviews with companies that already operate hydrogen trucks.

Findings

Key challenges to adopting hydrogen- powered heavy- duty trucks in Finland include insufficient refuelling infrastructure remote regions, uncertainty around fuel and technology costs, lack of financial incentives, and stakeholders' reluctance to bear risks associated with an emerging technology, and how early adopters have addressed these barriers.

Research implications

The study offers new insight into the specific challenges LSPs are facing and shows how similar barriers have been addressed in early adopter contexts. This adds detail to an area where empirical evidence is still limited.

Practical implications

The findings help logistics companies and policymakers identify the most pressing challenges to hydrogen truck adoption and show which measures from early adopter contexts could support infrastructure development, targeted incentives and stronger ecosystem collaboration in Finland.

Social implications

Successful adoption of hydrogen-powered heavy-duty trucks supports climate targets, reduces emissions, and strengthens regional competitiveness, contributing to UN SDG 13 (Climate Action). Original This paper bridges the gap between adoption intentions and real-world implementation, for overcoming barriers and leveraging lessons from the early adopters.

Paper 70 - Full Paper

Carbon accounting for scope 3 emissions: differentiating business activities through an activity systems perspective

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Keywords: Carbon accounting; Scope 3; Supply chain emissions; Activity system perspective; Corporate Sustainability Reporting Directive

Purpose

The Corporate Sustainability Reporting Directive (CSRD) obliges certain companies to conduct carbon accounting. To address the challenges of performing carbon accounting, particularly with regard to scope 3 or supply chain emissions, this study identifies the business activities behind the officially proposed carbon accounting.

Design/methodology/approach

The study is founded on the activity system perspective, and utilised empirical data derived from twenty in-depth interviews with key informants from German manufacturing firms. The Gioia Methodology was utilised to analyse the empirical data.

Findings

The analysis identified three activity systems for carbon accounting, which differ based on a firm's experience with carbon accounting. One such system is the one that has been identified to monitor the activities of firms that have initiated carbon accounting. The second system presents the activities of firms that engage in carbon accounting for three years, while the third system shows the activities of firms that have been conducting carbon accounting for more than five years.

Research limitations/implications

The findings demonstrate a focus on Germany and are applicable exclusively to firms obliged to the CSRD. Future research could investigate the transition paths and prerequisites between the systems.

Practical implications

The identified activity systems enable practitioners to benchmark their current carbon accounting activities and provide a framework to align or define activities in consideration of carbon accounting experience level.

Originality/value

By introducing three activity systems, this study advances the field of carbon accounting by offering a nuanced framework to account for scope 3 (supply chain) emissions based on accounting experience.

Paper 71 - Work in Progress

Mapping and assessing supply chain risks - developing a supplier risk management tool in the context of medium-sized firms

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Keywords: Supply Chain Risk Management; Supply Chain Visibility; Risk Assessment; Design Science

Full Paper

Purpose

Due to the high volatility in supply chains, risk management is important in the context of global supply chains. The growing number of disruptive events is creating new requirements for risk management. The purpose of the study is to develop a tool to optimise supply chain risk management.

Design/methodology/approach

Drawing on existing literature, the Design Science Research Approach was used to develop a supply chain risk management tool for the upstream supply chain. The tool was tested in practice.

Findings

The study provides a tool that maps all available supplier tiers and flows of goods on a geographical map. The test of the tool demonstrates its potential to assess and enhance risk management and to inform decisions regarding sourcing strategies

Research limitations/implications

Due to its complexity, the tool does not account for interactions between Tier-N suppliers, while limited network communication often restricts data availability, constraining comprehensive end-to-end risk assessment.

Practical implications

Due to lack of relevant supply chain data, the mapping and assessing of all tier-n-supplier-levels and risks cannot always be guaranteed. Therefore, it is recommended to optimize network communication and combine the relevant data with other risk data to facilitate supply chain risk management. The tool of this study supports the risk management.

Originality/value

The tool offers a novel approach to managing the risks associated with suppliers and their flow of goods, and to initiating specific risk management measures.

Paper 72 - Full Paper

From waste to resource: an industrial symbiosis perspective on wood residues in the supply chain of industrialized timber house manufacturers

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Keywords: Circular economy; Bio-based material; Resource efficiency.

The purpose of this paper is to identify where wood waste occurs in the supply chain of industrially manufactured single-family timber houses and to explore how these wastes could be utilized in alternative value-creating processes to increase circular resource use. The study adopts a qualitative, multi-actor research design. Empirical data were collected through site visits, observations, unstructured interviews, meetings, and document analysis involving industrial timber house manufacturers, upstream suppliers, waste management actors, and district heating companies. Wood wastes occur across all stages of the supply chain and are largely used for energy recovery practices. A meaningful share of wood wastes consists of untreated, near-virgin material generated higher up in the supply chain. Key barriers to higher-value use include limited volume consolidation, regulatory ambiguity regarding material classification, and competition with established energy recovery systems, creating system-level trade-offs between manufacturing and energy sectors. As an exploratory qualitative study, the findings are not statistically generalizable. Future research should quantify waste volumes, assess techno-economic feasibility, and examine regulatory interpretations of pre-consumer wood waste.

Practical implications

The results indicate a need for coordinated governance mechanisms, such as intermediary actors capable of aggregating, quality-assuring, and redistributing wood wastes across supply networks. Redirecting waste from energy recovery to material use may extend carbon storage but also affect energy systems, highlighting the need for using a systems perspective. The paper contributes a network-level explanation of why pre-consumer wood waste remains underutilized in circular applications, linking supply chain management and circular economy research through the waste hierarchy.

Paper 73 - Full Paper

CO2 data is the new oil: scope 3 emission reporting in the transport service triad

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Keywords: Scope 3; Resource Dependence Theory; Transport Network; Transport Service Triad; Sustainable Supply Chain Management

Purpose

The purpose of the paper is to explore the Scope 3 emission reporting process within the transport service triad, transport buyers, logistics service providers (LSPs) and transport operators, focusing on network dependencies and power. Design An empirical study that describes the Scope 3 reporting process, mapping the actor roles and resource dependencies, particularly emission data, within the triad. The methods include interviews, workshops, and content analysis of legislative documents and sustainability reports. An analysis, grounded in Resource Dependence Theory (RDT), is conducted to explore the patterns and understand the process.

Findings

The paper shows that emission data emerges as a strategic asset that redistributes power. Transport buyers hold strategic power through contractual demands, LSPs act as the middleman, and transport operators hold operative control over emission data, driving reporting pressure and creating data-related dependencies.

Research implications

This paper offers insights and strengthens theoretical integration of RDT with regulatory perspectives. It extends RDT further into the context of Scope 3 by showing how emission data operates as a critical resource shaping the transport service triad.

Practical implications

The findings offer practitioners with insight into frictions arising when meeting Scope 3 reporting requirements. By revealing how these issues shift dependencies across the triad, the paper supports clearer data-sharing routines and more accurate allocation of reporting responsibilities. Original The paper moves beyond a dyadic focus by applying a network perspective that reflects the multi-actor structure of the triad. Showcasing how Scope 3 reporting shapes organisational dependencies, addressing relational aspects often overlooked.

Paper 74 - Full Paper

Creating value with circular logistics services

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Keywords: Logistics Services; Value Proposition; Circular Economy; Service-Dominant Logic

Purpose

This research investigates how logistics services address value in circular economy (CE). In this regard, while extant research in logistics focuses on utilities and value-added solutions, CE literature emphasizes resource-oriented outcomes through circular business models. Since logistics is a critical enabler of CE, this leaves a theoretical puzzle: how can logistics services, traditionally treated as cost driven operational support, facilitate value creation in CE? This study addresses the "logistics-service-value" connection by examining how service offerings align with circular strategies to facilitate multi-directional resource loops.

Design/methodology/approach

Drawing on Service-Dominant (S-D) Logic, the study investigates how the concept of value is articulated across a heterogeneous set of CE actors. Insights into logistics service offerings that facilitate circular material flows emerged from fourteen semi structured interviews. The resulting data set was then analyzed through thematic coding, emphasizing circular strategies, service elements, value proposition content, and the processes enabling resource integration.

Findings

The findings conceptualize logistics service providers leverage operant resources such as specialized knowledge and skills to shape their value proposition and the resource utility at the optimum level.

Research limitations/implications

The paper extends S-D Logic to the CE domain, emphasizing that the "service" dimension of logistics is paramount for circular value creation.

Practical implications

For practitioners, the research provides a roadmap for how firms can redesign logistics from transactional coordination toward service oriented circular value orchestration.

Originality/value

This research fills a critical theoretical gap by integrating the often-overlooked "service" dimension into the intersection of logistics and the CE.

Paper 75 - Full Paper

Flexible in theory, constrained in practice: operationalising spatiotemporal flexibility in logistics networks

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Keywords: Spatiotemporal flexibility; Interorganisational power; Logistics networks; Physical flows; Information flows

Purpose

While flexibility is frequently highlighted as a key performance dimension in logistics, it remains conceptually elusive and often treated in isolation from the organizational and relational contexts in which it is enacted. The purpose is to increase the understanding of how flexibility is enabled or constrained through power in logistics networks.

Design/methodology/approach

The study adopts a qualitative case-based approach focusing on a logistics network involving four types of actors, a wholesaler being the most central. Empirical material consists of interviews, observations and workshops. The analysis is informed by a power perspective.

Findings

Preliminary findings show that flexibility varies over time and across actors and flows. Physical solutions enable route and delivery flexibility within delivery performance constraints, while the flexibility potential of information systems often remains unrealized due to coordination requirements in system design. Moreover, network actors may jointly resist specific forms of flexibility, such as intermediate storage, based on shared risk considerations.

Research limitations/implications

The study is limited to a single network context, and future studies can test the findings' applicability across different logistics settings.

Practical implications

Findings highlight the need for joint reflection among interdependent actors to identify what forms of flexibility are desirable and where in these should be located.

Social implications

Improved alignment between physical and informational flexibility has potential for more efficient transport planning, reduced handling, and lower environmental impact.

Originality/value

The paper contributes by conceptualising logistics flexibility as a relational, spatiotemporal phenomenon emerging from power balances across physical and informational flows.

Paper 77 - Full Paper

Insider research in logistics and supply chain management

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Keywords: Insider research; Logistics; Supply chain management; Methodological positioning

INSIDER RESEARCH IN LOGISTICS AND SUPPLY CHAIN MANAGEMENT ABSTRACT Full Paper

Purpose

The purpose of this paper is to frame insider research as a research approach for studying emergent and practice-driven phenomena in logistics and supply chain management (LSCM). Many LSCM problems are not given ex ante but are constituted through ongoing operational and strategic work. Research designs grounded in predefined constructs and episodic outsider access therefore tend to study problems after they have been defined, rather than their emergence. Insider research differs epistemically by enabling researchers to engage with problem formation as it unfolds within practice, thereby accessing forms of knowledge that remain largely unavailable to traditional outsider approaches.

Design/methodology/approach

The paper conceptually positions insider research alongside established in-depth research approaches in LSCM.

Findings

Insider research in LSCM enables access to emergent problems and forms of coordination that are difficult to capture through episodic and external research perspectives. Insider research supports a deeper understanding of how operational decisions and strategic priorities interact across organizational boundaries in everyday work. At the same time, insider research introduces tensions related to dual roles and analytical distance. Rigor in insider research depends on making the researcher's role explicit and documenting empirics and interpretations over time.

Originality/value

This paper contributes to methodological discourse by positioning insider research within the LSCM field. It moves beyond generic discussions of insider research in organization studies by specifying how it shapes access, problem formulation, and rigor in LSCM research.

Keywords

Insider research, Logistics, Supply chain management, Methodological positioning.

Paper 78 - Full Paper

Modeling logistics support with additive manufacturing of a mechanized battalion

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Keywords: Defence logistics; Military maintenance; Additive manufacturing; Inventory planning

Purpose

In attritional warfare, such as now in Ukraine, repair and maintenance operations (RMO) are crucial in battlefield success. In this paper we study Additive Manufacturing (AM) enabled RMO at the battalion level.

Design/methodology/approach

We build a discrete event simulation model of a mechanized battalion RMO for decisive battles of two weeks. The model is based on our previous military maintenance simulation studies with new features such as inventory transshipment at field-maintenance level, replenishment of metal parts from depot-level and inventory-AM tradeoff.

Findings

Previous findings without transshipment and replenishment from depot-level indicate that maintaining operative capability can only be achieved with low battle intensity and high initial inventories. Initial results suggest that battle intensity remains a problem while initial inventories can be allocated optimally.

Research limitations/implications

Validating the simulation model is understandably difficult and future battles including the damage model may differ from the one we have used, neither are all vehicle parts modelled. Generalisations are therefore subject to caveats.

Practical implications

Discrete event simulation modelling with best available input parameter values enables testing combinations of variables for the battalion RMO performance. Such an approach could be valuable for military planners.

Originality/value

This kind of defence maintenance logistics concept as such has not been previously modelled. Our findings will provide insights for novel RMO concepts for military planners.

Paper 79 - Work in Progress

The (hidden) role of software providers in the digital transformation of road freight transport

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Keywords: Software provider; Freight transport; Digitalization; Socio-technical systems theory; Case studies

Purpose

Road freight transport accounts for 70-90% of inland freight transportation in the Nordics, yet digital transformation lags behind other sectors. Prior research attributes the slow digitalization in road freight transport to organizational dynamics such as limited staff resources, digital illiteracy, low leadership support, and adherence to traditional working practices. These dynamics suggest that road freight organizations require software providers (SPs) that go beyond their role of technical enablers and actively support the organizational change process. This paper addresses the question: How do SPs influence the digital transformation in road freight transport organizations?

Design/methodology/approach

The study examines multiple cases of Scandinavian road freight transport organizations implementing core systems (i.e., transport management systems). The study uses the socio-technical systems perspective to analyse the role of the SPs. The dataset consists of interviews, observations, internal project documentation, and software descriptions.

Research limitations/implications

The case-based design makes the study context dependent; factors such as organizational size and prior digital experience may influence the results. The results provide analytical generalization, for statistic generalization further research needs to be conducted.

Originality/value

The study extends the emerging literature on organizational dynamics in digital transformation by uncovering a hidden role of SPs. The insights can enable SPs to better support freight transport organizations in addressing these organizational challenges, and thereby, accelerate digital transformation in the road freight transport industry.

Paper 82 - Work in Progress

Rethinking the supply chain risk management process - insights from a case study with remote sensing technology and machine learning analytics

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Keywords: Supply Chain Risk Management; Remote Sensing; Machine Learning; GeoAI; Digital technologies

Work in Progress

Purpose

Global supply chains are increasingly exposed to disruptions that evolve faster than traditional supply chain risk management (SCRM) processes can detect and respond to. Advances in machine learning (ML) enable new forms of data-driven risk analysis for SCRM. This study examines how the integration of remote sensing technologies, related data, and ML-based analytics is reshaping the established SCRM process. The study focuses on highlighting changes in established SCRM processes as well as the resulting implications for information processing in supply chains.

Design/methodology/approach

The study draws upon Information Processing Theory and Business Process Reengineering Theory in order to examine the shifts in the information processing capacity and the changes in the established SCRM process. The study employs a case study within the context of deforestation-related supply chain risks and geographical artificial intelligence methods. These methods combine remote sensing data and ML-based analytics.

Findings

As a work-in-progress paper, empirical findings are currently under development. However, initial results indicate that that the combination of digital technologies enable a shift from reactive to proactive risk management.

Practical implications

The study is expected to provide practitioners with a more profound understanding of how digital remote sensing technologies reshape information processing capacities and present a novel process for SCRM.

Originality/value

The study intends to propose a novel SCRM process that arises from the integration of digital technologies, specifically remote sensing technologies, which have the potential to transform the information processing capacities of firms.

Paper 83 - Work in Progress

Unfolding proactive approaches to the initiation of the manufacturing relocation process

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Keywords: Relocation; Reshoring; Decision-making; Proactive; Resilience

Purpose

This study explores how manufacturing companies develop and execute a proactive approach to initiating the manufacturing relocation process.

Design/methodology/approach

The study uses a case study research strategy, where data is collected using semi-structured interviews with company managers. The analysis is built on established concepts in the literature related to proactivity, strategic decision-making, and manufacturing relocation.

Findings

Preliminary findings show that decision-makers gain awareness of one or more driving factors for relocation that, through evaluations, trigger a proactive response to a relocation decision. Motivations for a proactive approach are protectionism, risk awareness and reducing ambiguity. The proactive approach is often developed overtime by the decision-makers.

Research limitations/implications

The research implies that manufacturing companies can proactively approach the manufacturing relocation decision from different perspectives and approaches. Establishing a proactive approach can enhance resilience and enable companies to maintain a competitive edge. Generalizability of the findings is limited by the sample size. Future research could investigate how proactive decision-making can enhance relocation decisions.

Practical implications

The anticipated contribution for practitioners is to gain a deeper understanding of how to incorporate a proactive approach to manufacturing relocation decision-making, promoting resilience. Moving from a reactive to a proactive approach enables decision-makers to adjust more quickly to changes in the external environment and adjust to internal performance.

Originality/value

In general, the understanding of proactive manufacturing relocation decision-making is lacking. Thus, this study expands upon an important and timely topic.

Paper 84 - Full Paper

Seaports' sustainability transition under rapidly changing external pressures

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Keywords: Sustainability; Seaports; Defense; NATO

SEAPORTS' SUSTAINABILITY TRANSITION UNDER RAPIDLY CHANGING EXTERNAL CONDITIONS ABSTRACT Full Paper

Purpose

This paper examines how rapidly changing external conditions shape the sustainability transitions of Swedish seaports, with particular attention to national defense requirements, Sweden's NATO accession, and escalating climate risks.

Design/methodology/approach

The study adopts a multiple case study design based on semi-structured interviews with seven strategically selected Swedish seaports. Based on the empirics, the paper develops a conceptual model illustrating how geopolitical and climatic shifts interact with ports' sustainability work and decision-making processes.

Findings

The findings show that ports' sustainability transitions are primarily shaped by governance arrangements and resource prerequisites. Several ports face structural funding constraints, where ownership models limit long-term investment and make sustainability initiatives dependent on external financing. Organizational capacity is a further limiting factor, as shortages of specialist staff and administrative resources constrain both strategic development and access to external funding. External pressures also vary geographically: in less defense-relevant ports, defense requirements are largely anticipatory, whereas in more defense-relevant ports they translate into ongoing operational demands that directly affect infrastructure planning. Although climate adaptation is widely acknowledged as necessary, it remains underfunded across cases.

Originality/value

The paper contributes a framework explaining how ownership structures, resource constraints, and security- and climate-related pressures jointly shape sustainability transitions in seaports. It highlights why transition pathways may stagnate despite available technologies and identifies leverage points for policy and organizational design to support sustainable and resilient port development.

Keywords

Sustainability, Seaports, Defense, NATO.

Paper 85 - Full Paper

Freight transport procurement platforms: evaluation of administrative and transport-related efficiency in digital platforms

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Keywords: Freight transport efficiency; Data sharing; Digital platforms; Load factor; Transport procurement

Purpose

The paper's purpose is to examine how the digitalization of transport procurement through digital platforms and the analysis of platform-generated data can contribute to increased efficiency in the freight transport system, including both administrative and transport efficiency.

Design/methodology/approach

The study is based on a qualitative research design using semi-structured interviews with transport buyers and transport providers operating within the industrial and retail sectors. All respondents use digital platforms for procuring and booking freight transport by road and other modes. In total, 9 actors were selected in collaboration with two platform providers. Data were collected through online interviews and analyzed using an iterative thematic analysis approach.

Findings

The findings show that digital platforms contribute to efficiency through reduced administrative workload, improved identification of backhaul opportunities, and increased focus on optimizing load factors. Transport buyers report reduced manual coordination, while carriers highlight improved visibility of potential return loads. However, geographical constraints and limited integration between systems hinder further efficiency gains. Data sharing between actors still requires substantial manual effort.

Research limitations/implications

The study is limited to a qualitative sample of actors connected to two digital platforms, which may limit generalizability.

Practical implications

Improved system integration, flexible time windows, support for part loads, and visualization tools can enhance planning and reduce empty running. Platform design plays a key role in unlocking unused transport capacity.

Originality/value

The paper provides novel empirical insights into how platform-based transport procurement and data use can enable freight transport efficiency without requiring direct data sharing between competing actors.

Paper 87 - Work in Progress

Decarbonization of inland container transport: a scoping review of chain governance and incentives alignment in door-to-door services

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Keywords: Door-to-door; Container shipping; Intermodal transport; Decarbonization; Inland container transport; Value creation

Purpose

The organizational, contractual, and governance mechanisms needed to integrate inland container transport decarbonization within the door-to-door shipping chain remain weakly understood. Addressing these gaps is essential to enable an integrated, low-carbon door-todoor container transport service.

Design/methodology/approach

This paper applies a five-stage scoping review to examine how academic research has addressed the decarbonization of inland container transport, with particular attention to chain boundaries, actors, and coordination mechanisms.

Findings

The review shows that inland container transport is often examined separately from ocean shipping, with unclear chain boundaries and underexplored incentive misalignments. The paper therefore proposes a framework identifying coordination failures and key actors and interfaces shaping decarbonized inland services.

Research limitations/implications

Future research should adopt a chain perspective capturing service design, development and execution; identify distinct market segments; and model incentive structures and costsharing configurations.

Practical implications

Decarbonized inland container transport services should be tailored to distinct shipper market segments, requiring industry actors to adapt offerings while providing inland operators with clearer and longer-term commitment signals.

Social implications

With global container flows projected to double by 2050, reducing emissions from the carbon-intensive inland leg is critical to achieving freight transport decarbonization.

Originality/value

The paper contributes to a chain-based perspective on inland container transport decarbonization, highlighting how its design and development are shaped by its embeddedness within door-to-door services.

Paper 88 - Work in Progress

Leveraging community participation in humanitarian AI: the role of co-design

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Keywords: Humanitarian supply chains; Artificial intelligence; Co-design; Community participation

Purpose

This paper explores how community participation is leveraged in humanitarian supply chain projects involving AI-based tools. Further, it outlines how community participation could be enhanced through co-design approaches.

Design/methodology/approach

We conduct a comparative case study of United Nations AI initiatives, using a qualitative analysis of primary data from interviews and secondary data from documents.

Findings

Fewer than 2% of UN AI initiatives are tools intended for use by the general public or civil society organizations in humanitarian settings. Among the six identified projects intended for community use, we find low to no community involvement across design, development, and implementation processes. Based on our findings, we demonstrate how co-design approaches would enhance community participation, including by providing contextual understanding to system design, identifying ethical issues, assessing the need for different tools, flagging potential data biases, diversifying development teams, supporting data collection and verification, and providing critical feedback to improve systems.

Research implications

The study extends research on humanitarian artificial intelligence by documenting the limited role of community participation in UN-led AI initiatives. It positions co-design as an analytical lens for examining how community participation influences data practices, ethical risk identification, and decision-making processes in humanitarian supply chains.

Practical implications

Co-design offers a path for advancing the humanitarian sector's commitment to localization, while reducing potential harm from AI tools in vulnerable settings.

Originality/value

While community participation has been gaining scholarly interest, this paper is, to the best of our knowledge, the first to explore the phenomenon in humanitarian technology development.

Paper 89 - Work in Progress

Social capital and socially responsible procurement in agricultural cooperative: evidence from indonesian coffee

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Keywords: Socially responsible procurement; Cooperatives; Qualitative comparative analysis; Smallholder coffee; Coffee supply chains

This paper investigates how coffee cooperatives in Indonesia respond to diverse, and sometimes conflicting, stakeholder expectations from farmers, buyers, certification bodies, governments, and NGOs when implementing socially responsible procurement (SRP). The purpose is to identify the causal mechanisms and configurations through which cooperatives adopt, partially adopt, or struggle with SRP in smallholder-based sourcing contexts. The study adopts a multiple-case research design combining an inductive qualitative approach with a configurational analytical technique. First, Gioia-based qualitative coding is used to explore how cooperative leaders make sense of stakeholder demands and SRP requirements. Second, Qualitative Comparative Analysis (QCA) is applied to examine how combinations of stakeholder pressures, cooperative governance capabilities, and contextual conditions jointly shape SRP outcomes. This approach is grounded in a neo-configurational perspective that recognises causal complexity and conjunctural causation. The analysis is expected to identify multiple, equifinal causal pathways through which cooperatives navigate stakeholder demands. SRP outcomes are shown to emerge from interacting mechanisms related to stakeholder influence, governance, and capability development, as well as from interpretive framing by cooperative leaders, rather than from a single dominant driver. The study focuses on cooperatives within the Indonesian coffee sector, which may limit generalisability. The findings offer insights for cooperatives, buyers, and certification bodies on how governance structures and adaptive strategies can be aligned to manage stakeholder tensions more effectively in SRP implementation. The paper advances SRP research by integrating sensemaking and configurational analysis to explain how cooperative-led procurement practices emerge under multi-stakeholder pressures.

Paper 92 - Full Paper

Dynamics of community participation in humanitarian supply chains during conflict

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Keywords: Humanitarian supply chain; Community participation; Complex Adaptive Systems; Emergency response; Conflict

Purpose

The study conceptualizes community participation in humanitarian supply chains during conflict as a dynamic phenomenon, identifying the factors that enable or constrain it over time.

Design/methodology/approach

The escalation of the war in Ukraine (2022–2025) is examined as a qualitative case study, drawing on primary interview data with local staff and secondary data from international humanitarian organizations.

Findings

Three distinct patterns of community participation were identified, depending on the stage of conflict and interactions with other actors. During the first three months, community actors were responsible for virtually all functions related to the distribution of humanitarian aid throughout the country. In the following months, the space for these actors was reduced as international organizations entered the humanitarian supply chain. Later, as funding and training became more accessible, community-based organizations and groups reassumed responsibility for selected functions, particularly at the last-mile and beneficiary-interface levels, operating alongside international actors rather than independently.

Research limitations

The analysis focuses on a single conflict setting characterized by relatively high pre-existing community participation. Future research should examine other conflict and disaster contexts for comparison.

Practical implications

Viewing community participation as dynamic can support the design of more inclusive humanitarian supply chains, advancing localization objectives while reducing costs and expanding reach.

Originality

This is the first empirical paper to address community participation as a dynamic phenomenon, analyzing how it evolves in interaction with other players in the humanitarian supply chain, through the lens of Complex Adaptive Systems.

Paper 93 - Work in Progress

Understanding freight network resilience through the resilience funnel

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Keywords: Freight transport; Supply chain resilience; Transport networks; Resilience funnel; Disruptions

Purpose

Freight transport disruptions increasingly threaten the continuity of supply chains, yet resilience research often treats transport networks and supply chains as separate analytical domains. This study addresses this gap by examining transport network using the supply chain resilience funnel as a framework.

Design/methodology/approach

The study draws on four qualitative case studies of disruptive events across various modes of transport. The data includes semi-structured interviews of supply chain actors impacted by the disruption, and secondary sources, such as news articles. The cases capture disruption preparedness, response, and adaptation, based on comparative analysis of multiple actors and freight flows.

Findings

The findings show that transport networks shape the initial width of the resilience funnel by enabling or constraining available supply chain response options through infrastructure characteristics, institutional arrangements, and capacity conditions. As disruptions unfold, supply chains activate a progressively narrower set of options, such as rerouting, modal shifts, stockpiling, production rescheduling, and sourcing adjustments, reflecting the funnelling of resilience choices under growing constraints. Importantly, actions taken at the supply chain level can also influence transport network functioning, illustrating reciprocal and cascading resilience effects.

Originality/value

This study contributes to theory by extending the resilience funnel framework to the freight transport context, highlighting resilience as a multi-level, processual phenomenon. Managerially, the findings emphasise the need for coordinated resilience planning across transport networks and supply chains to sustain freight flows during disruptions.

Paper 99 - Work in Progress

Integrating ship design into corridor network design for co2 shipping: a case study of the Sweden–northern lights chain

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Keywords: CO2 shipping; Carbon capture and storage (CCS); Transport network design; Freight transport; Mixed-integer linear programming

Carbon capture and storage (CCS) is increasingly recognised as a necessary component of net-zero strategies for hard-to-abate industries, yet large-scale deployment remains constrained by the cost and reliability of CO2 transport. In practice, ship design for CO2 transport and logistics network design are often treated as separate problems: vessel characteristics are optimised in isolation, while transport planning assumes a given fleet when designing corridors, hubs and service patterns. This separation risks misaligned design choices that inflate costs and constrain operability.

We propose a corridor-level network configuration model that integrates ship design decisions directly into transport network design for the Sweden–Northern Lights CCS chain. The model is formulated as a mixed-integer linear programme linking Swedish industrial capture sites, optional intermediate hubs and a storage terminal to a discrete set of candidate ship classes. Ship class selection and fleet size are endogenised, while operability is enforced through corridor-specific constraints on ship dimensions, draught limits, port times and annual ship-hour budgets. Costs are computed for sailing and port operations, and transport-related emissions are reported as key performance indicators.

Because captured CO2 has negligible market value at the gate, the transport system must deliver reliable service at minimum unit cost. Model outputs include cost per tonne, fleet composition and utilisation, routing and transhipment patterns, storage requirements and emissions per tonne transported, supporting coordinated ship design, transport procurement and port planning.

Paper 100 - Full Paper

Applying decoupling thinking to patient flow: a case study of specialist-primary care handovers

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Keywords: Customer order decoupling point; Service supply chain; Hospital municipality interface

Despite growing interest in decoupling as a flow-design strategy in healthcare, we lack empirical understanding of how instability in the conditions of frail older patients with complex needs shape patient flow at handovers in multi-provider service supply chains. This paper examines how decoupling in specialist–primary care handovers is shaped by the characteristics of frail older patients.

We conducted a deductive, exploratory single-case study of frail older patients moving across a Norwegian specialist–primary care interface, with patient flow as the unit of analysis. Empirical material comprises interviews, workshops and electronic patient journal metrics. Data were analysed using the Gioia methodology.

The hospital–municipality discharge is the key decoupling point, where destination, service content and timing are late-bound. Long-term care capacity shortages create queues of discharge-ready patients. The municipality mitigates this by fast-tracking temporary services (short-term placement/enhanced home care), substituting hospital waiting with temporary capacity while postponing final allocation until shared decision-grade information and capacity is available. Frailty-related instability and information insufficiency limit advance planning, reinforcing late binding in a pull-triggered, push-sequenced flow.

The findings invite research to further explore the relationships between buffer substitution, information readiness, split decision rights and discharge-to-service lead times and variability across settings.

The study extends decoupling thinking for inter-organisational patient pathways by specifying buffer substitution, information readiness and split decision rights as core handover mechanisms when frail older patients with extensive and complex care needs are the flow object. It offers design principles for hospital and municipal managers seeking to reduce discharge queues and discharge-to-service lead-time variability.

Paper 101 - Work in Progress

Reference-dependent choice modeling of consumer preferences for sustainable delivery services

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Keywords: Urban freight; Last-mile delivery; Consumer behavior; Willingness to pay (WTP); City logistics

The rapid growth of e-commerce and on-demand platforms has transformed urban freight from a supply-push system into increasingly consumer-driven. While logistics operations have been frequently studied, consumer decision-making is still commonly treated as aggregated, limiting the integration of behavioural demand formation with delivery-supply responses. This study addresses this gap by proposing an urban freight framework that considers consumers as active agents and examines its viability. The study combines large-scale consumer survey results from five European countries with a feasibility-validation framework. Consumer preferences and willingness to pay are used to estimate choice probabilities of differentiated delivery service packages. The outcomes are then linked to performance indicators to assess economic, operational, and reliability feasibility. The analysis shows that consumers exhibit heterogeneous and systematic preferences for delivery options, which can alter the demand pattern when multiple services are offered without compromising basic service reliability. So differentiated delivery offerings with pricing or incentive mechanisms have the potential to improve environmental and operational performance. The study relies on stated preference survey data, and operational costs are represented through simplified approximations. Future research could integrate revealed transaction data or network-level optimization to further validate the framework. The results provide guidance for e-commerce platforms and LSPs on delivery portfolio design to better align consumer preferences with sustainability objectives, encouraging shifts toward less resource-intensive delivery options while maintaining quality. This study contributes a novel perspective by framing urban delivery design around end consumers' choices, providing value to practitioners and policymakers seeking behaviour-informed pathways toward sustainable urban freight systems.

Paper 104 - Full Paper

From purpose to preparedness: corporate purpose and sustainability capabilities in supply chains

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Keywords: Corporate Purpose; Supply Chain Sustainability; Purpose Maturity; Sustainability Capabilities; Supply Chain Resilience

In increasingly volatile and sustainability-sensitive supply chain environments, Corporate Purpose has gained attention as a strategic orientation that goes beyond profit maximization. While prior research suggests that Corporate Purpose may influence organizational priorities, this study examines how different levels of purpose integration relate to sustainability action and the ability to address challenges in supply chains.

Despite growing interest in Corporate Purpose, supply chain management research lacks clarity on the mechanisms through which purpose becomes operationally relevant. It remains theoretically unresolved whether varying degrees of purpose integration are associated with systematic differences in sustainability practices and capabilities, or whether Corporate Purpose primarily remains a symbolic orientation.

The study draws on survey data from 202 professionals across manufacturing, logistics, and retail industries. Corporate Purpose maturity is operationalized through a three-group classification (Observers, Starters, Heroes). Correlation analyses, regression models, and group-based comparisons are applied to examine differences in sustainability measures, challenge management, and perceived preparedness.

The findings reveal a clear maturity gradient. Organizations with a highly embedded Corporate Purpose engage more strongly in concrete sustainability measures and report a higher capability to cope with sustainability-related challenges, particularly with regard to strategic prioritization. The study contributes to SCM literature by demonstrating that Corporate Purpose functions as a capability-enhancing orientation.

For managers, the results highlight Corporate Purpose as a strategic lever to strengthen sustainability implementation and decision-making under uncertainty. From a societal perspective, purpose-driven organizations may contribute to enhanced social and environmental sustainability and increased resilience in supply chains.

Paper 107 - Work in Progress

AI adoption in production logistics under constraints – policy development for efficient implementation while assuring security and privacy. Wippaper

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Keywords: Production logistics; AI; Autonomous mobile robots; Security ethics and privacy

With exponential artificial intelligence (AI) growth, the EU target to reach 75% AI implementation in industry by 2030. One AI-development area with significant potential, is in production logistics and autonomous mobile robots (AMRs), where AI can enhance performance and reduce risks. With AI, AMRs can evolve from knowing where and how to move, into capability to understand and avoid obstacles in the path. However, such developments incur challenges, such as dependence on data which has legal, ethics and business restrictions, lack of personnel trust in the adherence to ethics/regulations, and misalignment of policies and technology. Addressing these issues, the study aims to answer: How can policies be designed for collecting and handling AI systems data in production environments, enabling effective data use while ensuring information security and personal privacy? Four interviews and a workshop were conducted to identify barriers that slow down AI implementation. One barrier was lack of trust and understanding of how to implement AI systems without compromising personnel integrity, product quality, and information security. Further interviews with union representatives, technicians, HR representatives and legal advisors are planned to investigate specific concerns and how to ensure that the systems implemented are safe to use in all these respects, especially for AI as implemented in AMRs. This paper contributes with empirical understanding of AI-application and usefulness in production logistics. A managerial implication is the strategic importance of how to implement AI at lowest risk and highest gain. AI implementation requires ensuring profitable, safe and ethical applications.

Paper 109 - Work in Progress

Four nations face-off: on diversity, not on the ice

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Keywords: Supplier diversity; Sourcing; Similarity-attraction; Social sustainability

The paper explores pros and cons of supplier diversity (SD) from an organizational perspective, across several national contexts. Essentially, SD extends workforce / employee diversity (ED) into the upstream supply chain. There is a rich history of SD initiatives, starting in the United States in the 1960s, inspired by the Civil Rights Movement. Today, DEI programs are under pressure. During his first month back in office, President Trump issued Executive Order 14151, "Ending Radical and Wasteful Government DEI Programs and Preferencing," seeking to end DEI programs and offices across the federal government. According to a Fact Sheet from the White House, these executive orders terminate "diversity, equity, and inclusion" (DEI) discrimination in the federal workforce, and in federal contracting and spending. Interestingly, the private sector backlash on diversity began several years before 2025.

The research draws on several theoretical perspectives on diversity, including "zero-sum" vs. "win-win" assumptions, the diversity/unity paradox, and similarity-attraction versus social integration. Session attendees will have the opportunity to discuss these theoretical perspectives, share their experiences with diversity programs, and brainstorm on how to measure diversity benefits and challenges. Recent research results on SD pros, cons and pathways are included, focusing on the context and status of supplier diversity across six nations in North America (Canada and the United States) and Nordic countries (Denmark, Finland, Norway and Sweden). The intent of SD is to open doors of opportunity to all suppliers, irrespective of various personal characteristics (i.e. the diversity) of folks who own and manage the entities.

Paper 111 - Work in Progress

When plans break down: agentic AI and multi-actor coordination in disrupted supply chains

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Keywords: Supply Chain Coordination; Agentic-AI; Human-AI Collaboration

Coordination in disrupted supply chains often occurs under time pressure, information delays, and conflicting priorities across organizational boundaries. When planned sourcing and logistics decisions become infeasible, multiple groups of decision-makers must re-coordinate interdependent tasks and scarce human resources using partial information. Classical operations research approaches, including job-shop scheduling, support efficient task allocation under assumptions of stable information and centralized control, but provide limited insight into coordination shaped by interpretation, negotiation, and distributed authority.

Prior research on AI–human collaboration largely focuses on dyadic human–AI interaction. This paper extends that perspective to multi-actor coordination settings in which several human groups coordinate through a shared AI-mediated coordination interface. Agentic AI is defined as a bounded coordinating agent that aggregates distributed inputs, represents task dependencies, and proposes task allocations under human oversight.

The contribution of this paper is to establish conceptual foundations for future agent-based simulation by identifying coordination dependencies not represented in classical scheduling models. Drawing on Coordination Theory and Adaptive Structuration Theory, the study combines a structured literature review with exploratory expert interviews within the domain of supply chain coordination. Preliminary insights suggest that AI-mediated coordination can reduce coordination latency when information fragmentation dominates, but that outcomes depend on whether actors trust the AI system to represent sourcing and logistics dependencies accurately, on the reliability and explainability of its outputs, and on whether training data reflects cross-organizational coordination practices. The resulting framework clarifies key coordination dependencies and boundary conditions relevant for future agent-based studies of AI-mediated supply chain coordination.

Paper 115 - Full Paper

When you fail, you fail greatly: a multiple-case study of industrial internet of things platforms

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Keywords: Industry 4.0; Industrial Internet of Things; Failure framework; Cross-case study

Purpose

Industrial Internet of Things (IIoT) platform initiatives regularly fail, despite significant investments and their transformative potential to enable Industry 4.0 transformations.

Design/methodology/approach

Drawing on a six-factor failure framework, a cross-case study including five prominent IIoT platforms, namely GE Predix, IBM Watson IoT, Cisco Smart Cities, Ericsson IoT Accelerator, and SAP Leonardo was conducted. Utilizing directed content analysis of public archival sources, empirical evidence was mapped to predefined failure categories. These patterns were further interpreted through theoretical lenses of Diffusion of Innovation, Dominant Design, and the Technology Acceptance Model, providing a multi-faceted explanation for the observed outcomes.

Findings

Our findings reveal that Lack of Customer Value, Bad Ecosystem Fit, Product-Platform Confusion, and Premature Scaling emerge as the most dominant and recurrent patterns across the cases, and most of these IIoT platforms initiatives failed greatly in this respect. Practical implications (if applicable): The six-factor failure framework serves well as an effective diagnostic tool for assessing the failure risk of IIoT platform initiatives. In sum, a sequential approach is recommended: begin by solving specific, high-value customer problems using prevailing standards and hyperscaler infrastructures, demonstrate clear and auditable returns, and only thereafter expand into a broader platform ecosystem.

Originality/value

This research contributes to the literature on technology adoption and platform strategy by validating and operationalizing a failure framework in the complex context of Industry 4.0 transformation.

Paper 116 - Work in Progress

Humanitarian space: the final frontier

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Keywords: Aid workers; Attacks; Humanitarian logistics; Security

According to the Aid Worker Security Database (AWSD), major attacks on aid workers are at an all-time high. The humanitarian space seems to be shrinking. This renders the work of humanitarians, including logisticians, ever more challenging. While delivery, quality and cost remain important considerations among humanitarian logisticians, security is becoming increasingly critical. Drawing on secondary data, AI and archival research, this paper classifies major attacks on humanitarian aid workers and analyzes the responses of various aid organizations to these attacks. This paper focuses on all incidents reported in the AWSD during 2024.

In the face of such danger, aid agencies have a variety of security alternatives, from avoidance of or withdrawal from certain theatres of engagement to proactive strategies, such as embedding their activities and personnel with armed forces. Each of these alternatives comes with advantages and disadvantages. In addition, while the need for aid seems ever-increasing, donations may be declining and likely to come with "strings attached." Unfortunately, there are many opportunities for research in the area of humanitarian logistics security. Through advocacy and a broader approach to logistics, aid agencies may be able to enhance security. There is an opportunity to enlarge the "humanitarian space" – and increase security for aid workers and civilians, especially in areas of armed conflict.

Paper 117 - Work in Progress

You get what you sample: a comparative assessment of container freight rates

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Keywords: Container shipping; Freight rate analysis; Granger Causality

Purpose

Container freight rate analysis got a big topic with a steady raising number of articles being published. Unfortunately, most studies concentrate only on a few freight rate indicators like the Shanghai Container Freight Index (SCFI), despite nowadays, much more freight rate indices or benchmarks are available that may better reflect developments in container freight markets and so may be more applicable for benchmarking, forecasting, and contract negotiation purposes.

Design/methodology/approach

Ten public available container freight indices or benchmarks are subject to a thoroughly comparative assessment followed by exploration of their similarity in terms of temporally relatedness by means of Granger Causality testing.

Findings

Freight rate indices or benchmarks differ mainly because of their underlying sampling procedures that in turn provides different views on the complex freight rate mechanisms. This also explains well heavy deviations from one each other when freight rate hikes occur in times like the pandemic and red sea crisis. Research limitations/implications (if applicable) Given availability of historical backlog data, only main trade lanes since 2020 between China/FarEast, Europe and US East/West Coast could be assessed. Practical implications (if applicable) Researchers and practitioners are well advised to show more scrutiny when selecting a suitable container freight rate index or benchmark for their purposes. Societal implications (if applicable) ...

Originality/value

This a first comparative assessment of container freight indices/benchmarks.

Paper 122 - Full Paper

Toward greater efficiency in the flow of military public procurement. A case study in the spanish navy.

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Keywords: Defence procurement; Acquisition; Public procurement law; Contractual flow; Spanish Navy

To date, research on defence procurement has focused primarily on large-scale specific acquisition programs or those that impact essential national security interests, and the underlying causes and problems to support these military capabilities. Hence, the research gap addressed in this article focuses exclusively on the analysis of the military public procurement flow regulated by the Public Procurement Law, which applies to more than 80% of the Spanish Ministry of Defence's annual budget, a wide percentage that deserves scientific research. The main objective is to identify the main factors affecting the long lead times required in public procurement, finding viable solutions for redesigning the contractual flow, improving policy and decision making through research and analysis, drawing insights methodologically from primary (Delphi study conducted with practitioners - officers, non commissioned officers and selected civilian personnel - from all echelons and departments involved in procurement at the Spanish Navy's Logistics Arsenal in Cartagena), and secondary sources (literature review). The results will demonstrate that systemic measures and broader cultural and organisational change are needed all along the military contractual flow, focusing mainly in maintaining a professional workforce and the rational use of new technologies. The conclusions and proposed lines of work will identify the main challenges and contribute to optimizing the unstable balance between ensuring the necessary transparency and proper use of public funds with meeting the needs of military units in a timely and appropriate manner, thereby increasing the effectiveness and operational capacity of the Armed Forces, the ultimate goal of military contracting.

Paper 125 - Full Paper

Simulation gaming as a tool for conceptual learning in supply chain management

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Keywords: Supply chain management learning; Simulation gaming; Conceptual understanding

The importance of supply chain management (SCM) education has increased to a great extent over the last decade due to issues like sustainability awareness and various global disruptions. Students should be equipped not only to understand the causes of these issues and disruptions but also their dynamic effects. As supply chains involve complex interdependencies, education faces the challenge of helping learners develop a conceptual understanding of cause–effects. Conceptual understanding enables evaluating trade-offs and make informed decisions. It supports transferable competencies that go beyond operational skills. This study examines how a simulation game enhances conceptual understanding of dynamics in SCM. Data were collected from eight courses applying a simulation game in teaching supply chain dynamics at a Nordic business school. Over 700 students participated, and 292 completed both pre- and post-course surveys. Students were asked to assess, whether SCM dynamics related claims were true or false. Students' comprehension of these dynamics was analyzed by comparing responses before and after gameplay.

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